

Morphology of Maxillary Premolars

Dr Saba Arshad (Senior
Lecturer)

Department of Oral Biology



By the end, you should be able to...

01

Label the major crown and root landmarks.

02

Describe each tooth from five anatomical views.

03

Differentiate first from second premolars.

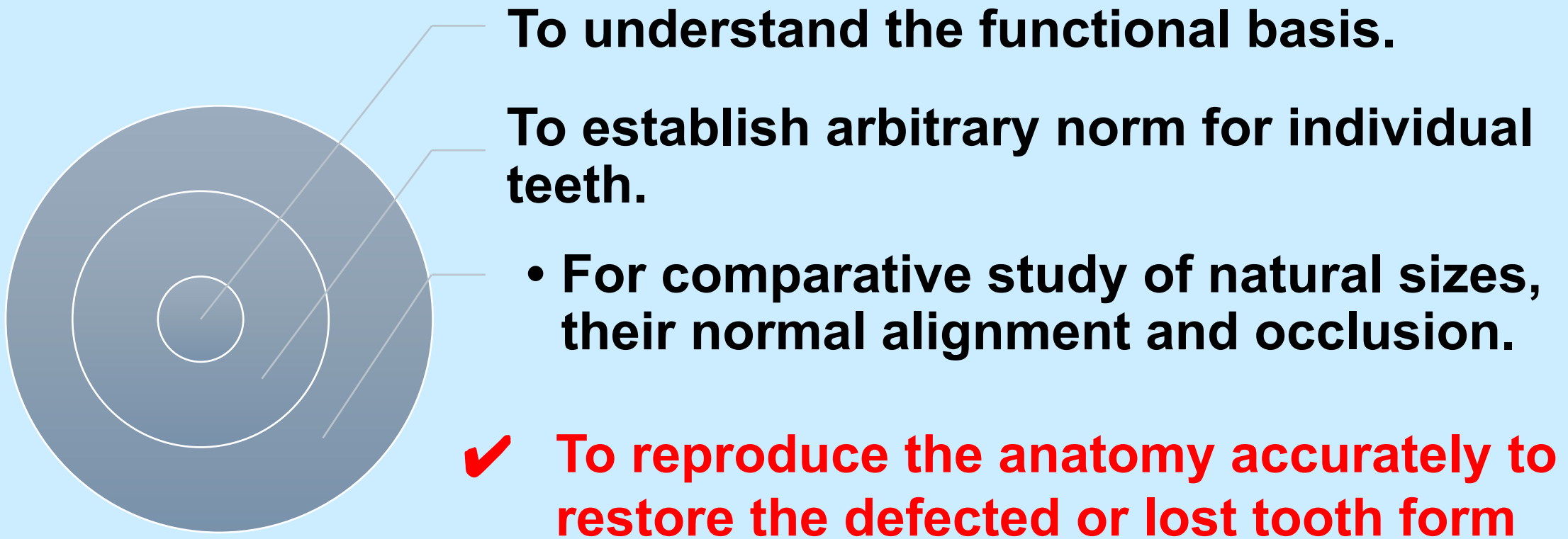
04

Relate form to function and clinical relevance.

05

Apply identification rules to unknown specimens.

Why it is important to describe every curve and segment of normal tooth





Premolars bridge tearing and grinding

Four maxillary premolars sit behind the canines and immediately in front of the molars.

4 developmental lobes

The lingual lobe forms a true lingual cusp.

Buccal + lingual cusps

Marginal ridges become part of the occlusal table.

First premolar

Sharper buccal cusp assists tearing.

Second premolar

Blunter cusps make grinding more efficient.

Maxillary arch

3rd
molar

2nd
molar

1st
molar

2nd
pre-
molar

1st
pre-
molar

2nd
molar

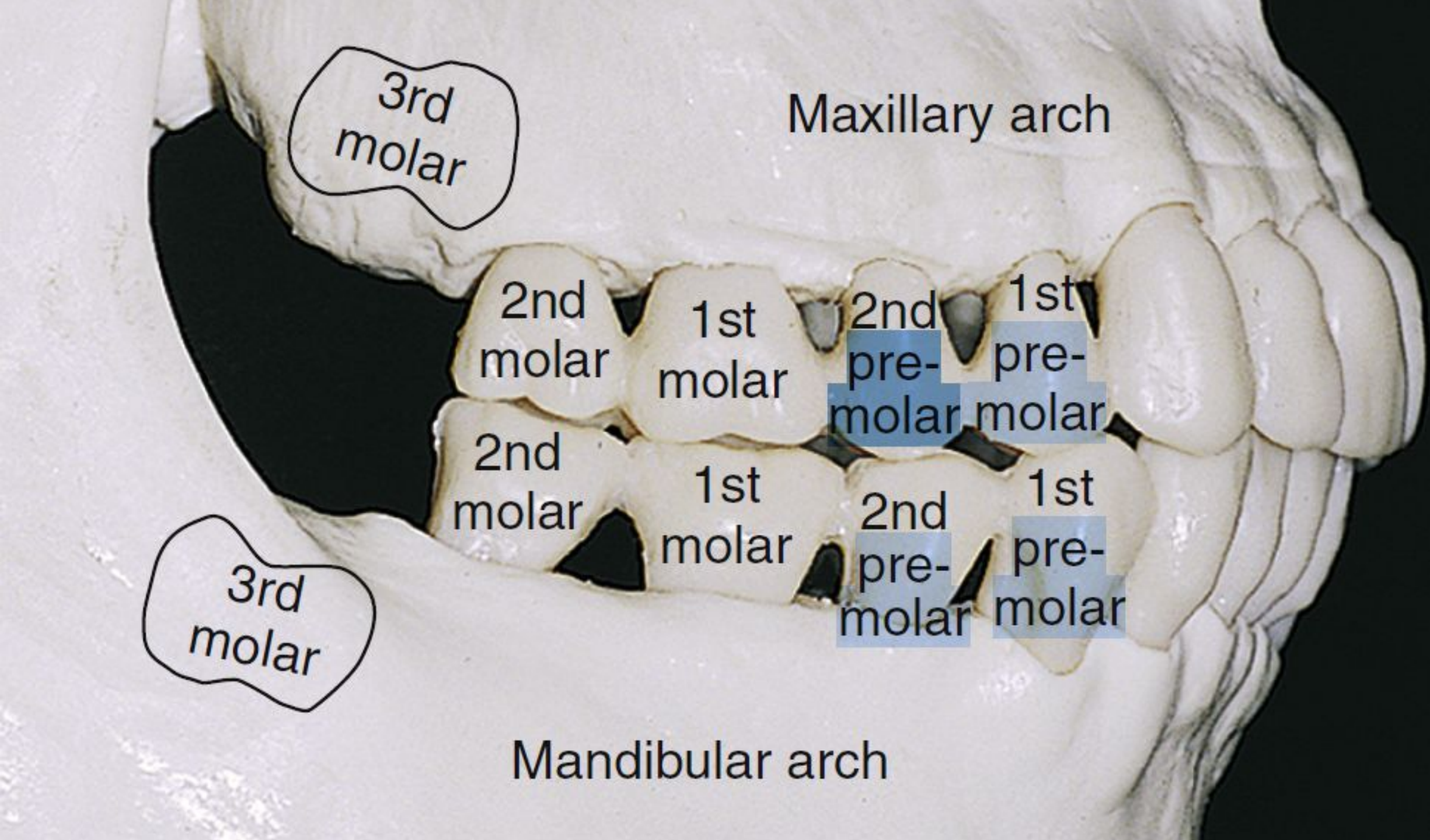
1st
molar

2nd
pre-
molar

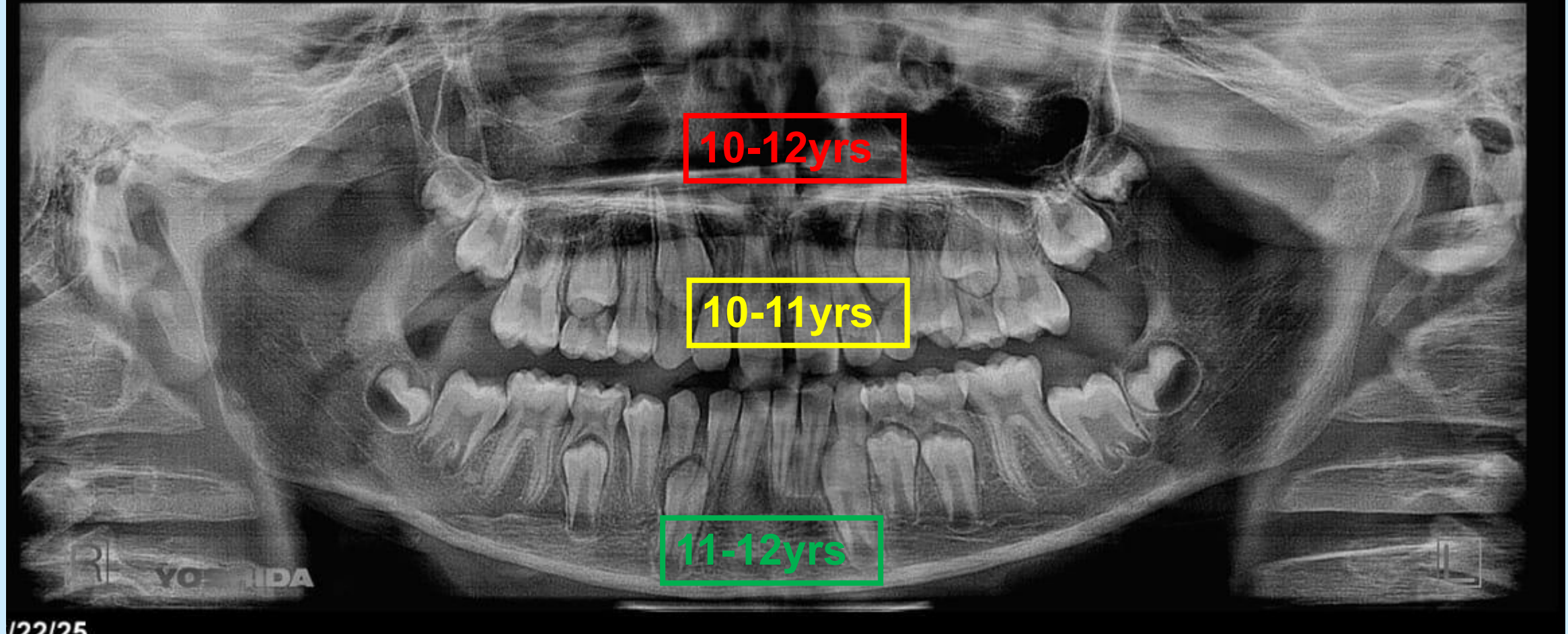
1st
pre-
molar

3rd
molar

Mandibular arch



Scale: 100%

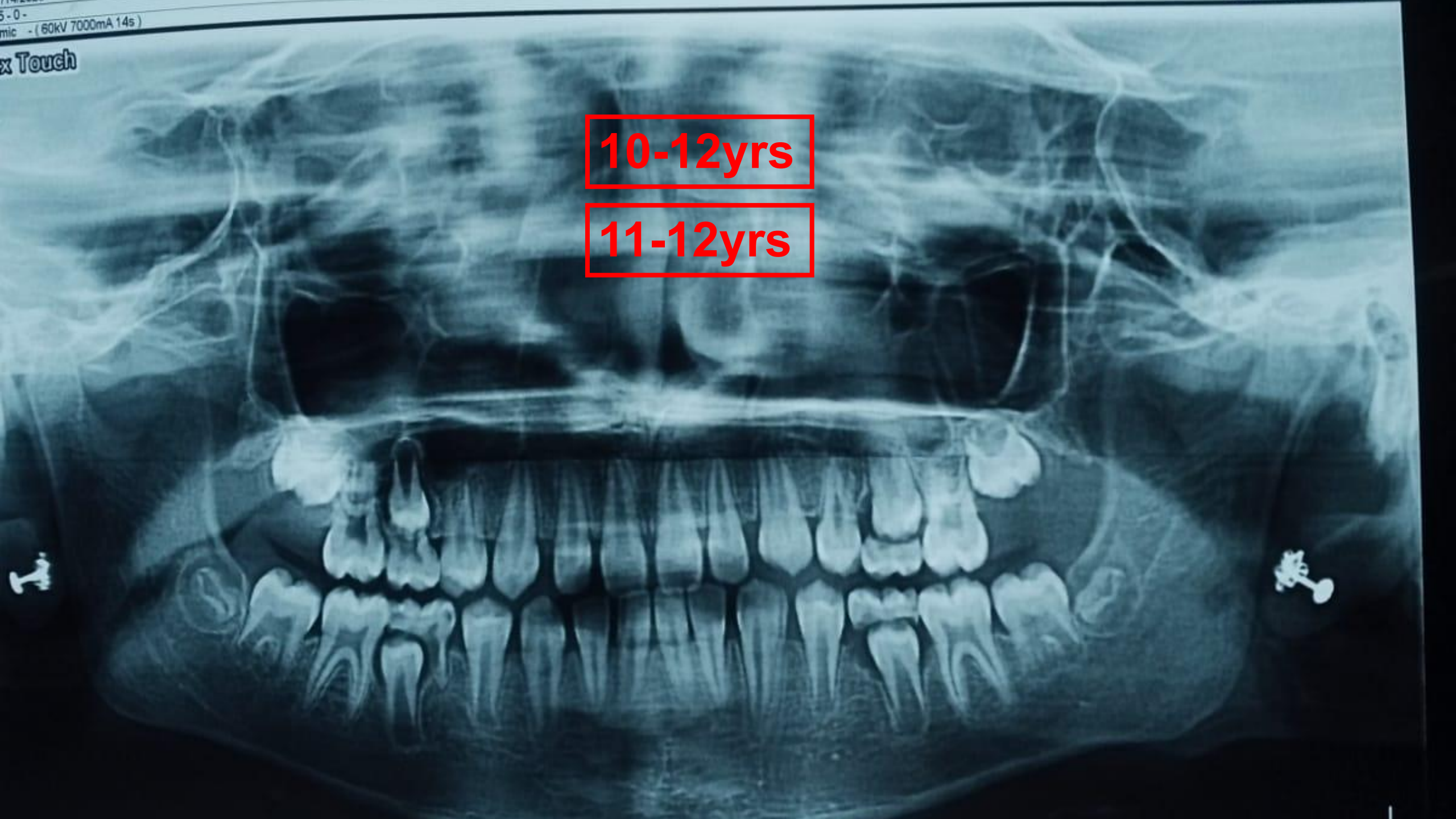


Identify dental chronological age of individual?

CHRONOLOGY OF THE PERMANENT SUCCEDANEOUS DENTITION

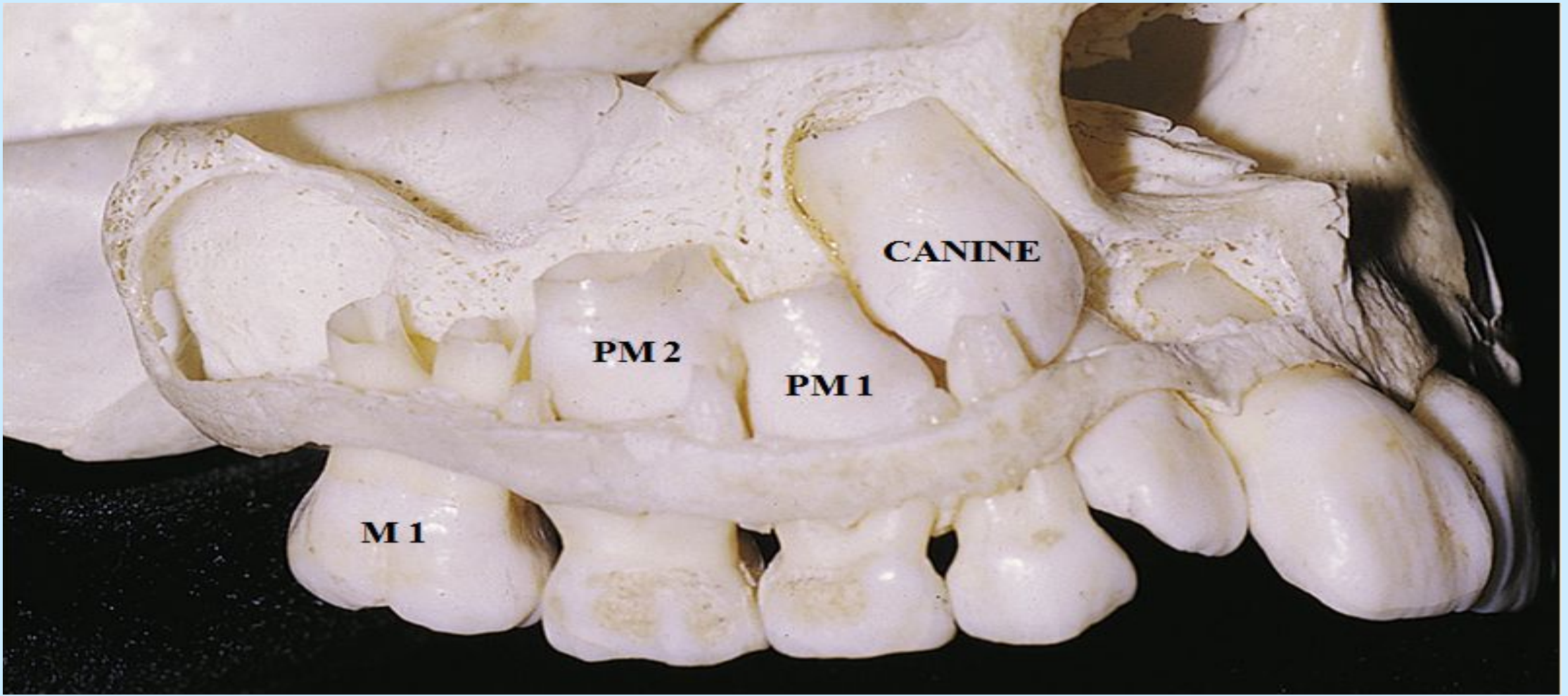
		1st Evidence Of Calcification	Crown Completed (Years)	Eruption (years)	Root Completed (Years)
Central Incisor	Max.	3-4 months	4-5	7-8	10
	Mand.	3-4 months	4-5	6-7	9
Lateral Incisor	Max.	10-12 months	4-5	8-9	11
	Mand.	3-4 months	4-5	7-8	10
Canine	Max.	4-5 months	6-7	11-12	13-15
	Mand.	4-5 months	6-7	9-10	12-14
1st Premolar	Max.	1½–1¾ years	5-6	10-11	12-13
	Mand.	1¼–2 years	5-6	10-12	12-13
2nd Premolar	Max.	2–2¼ years	6-7	10-12	12-14
	Mand.	2¼–2½ years	6-7	11-12	13-14

4-5 yrs

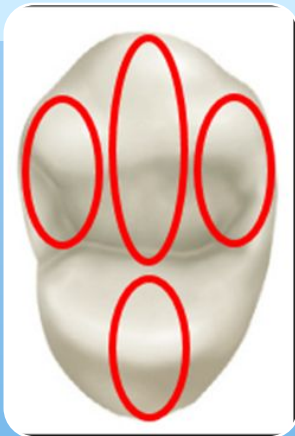


10-12yrs

11-12yrs

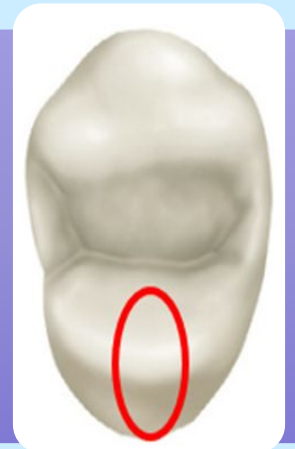


Anterior to molars; replace deciduous molars
(succedaneous Teeth).



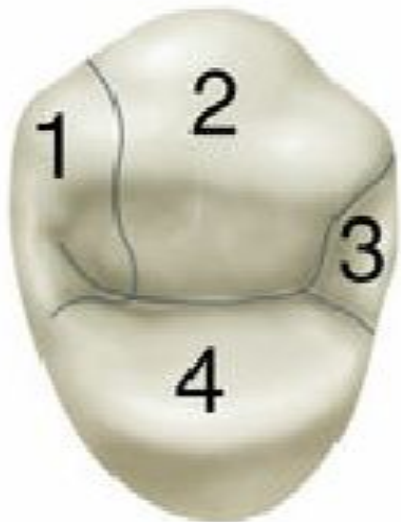
No. of developmental lobes:

- 4 lobes (Mesial, Buccal, Distal, Lingual). (same as anterior teeth)



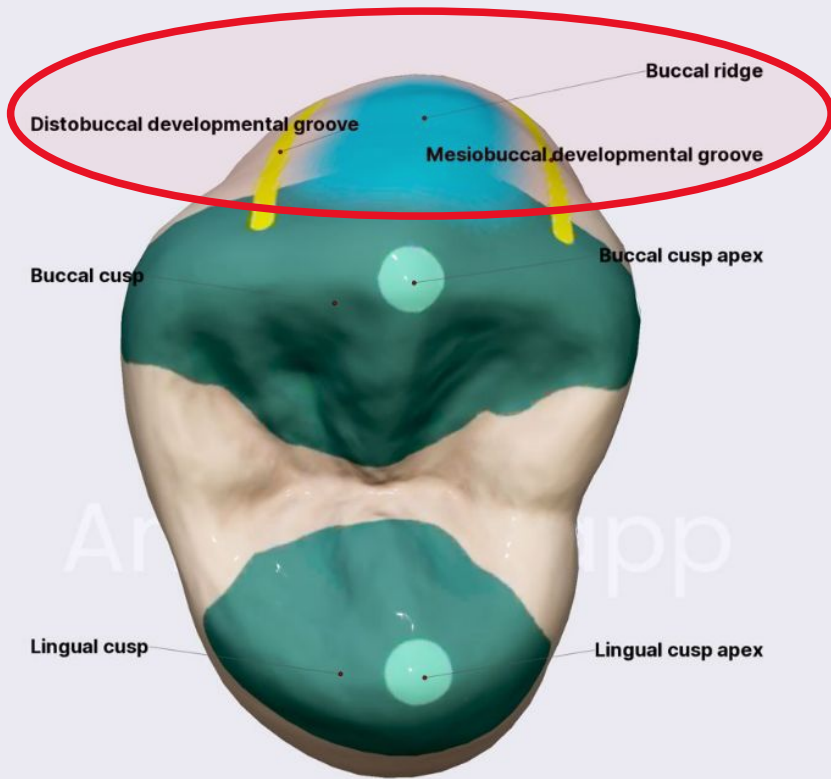
Fate of lingual lobe

- Well developed lingual cusp instead of cingulum in anterior teeth

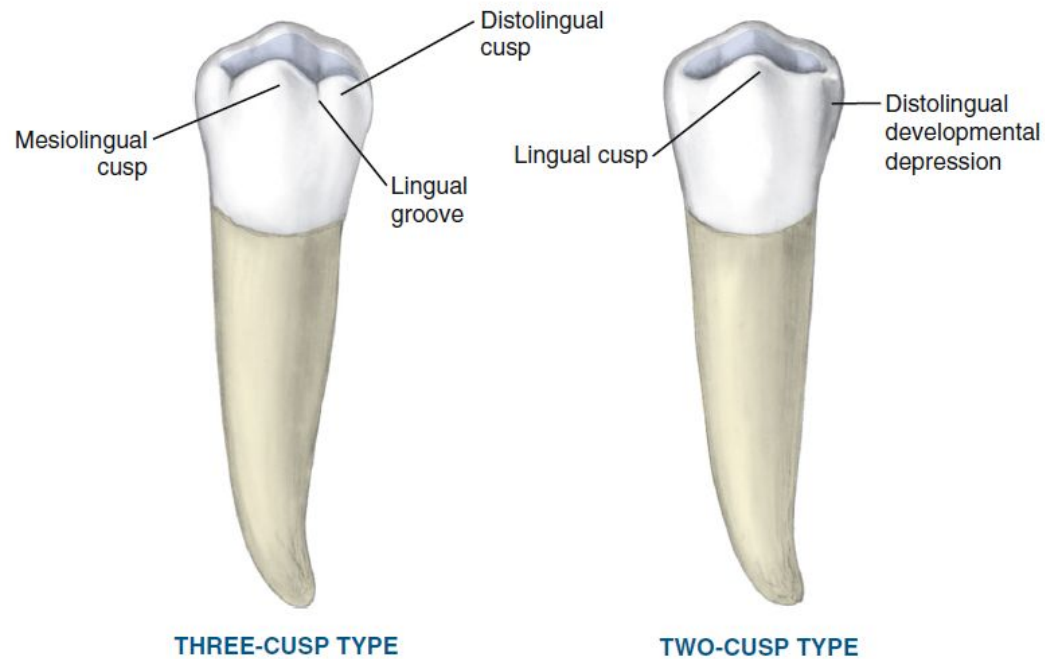


Fate of middle buccal lobe

- Remains well developed like middle labial lobe of canines

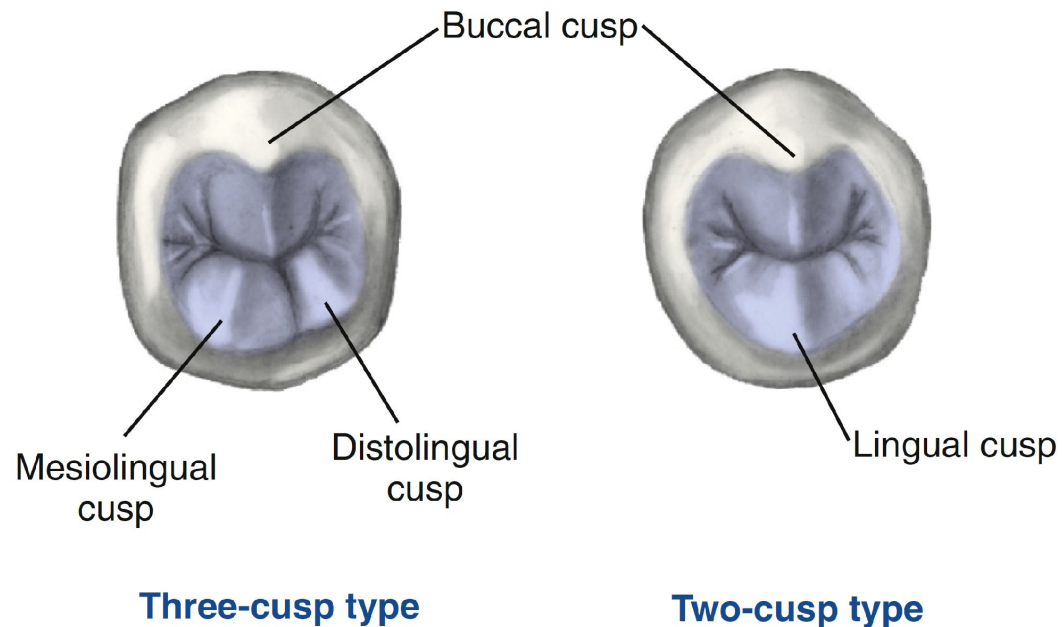


- The middle facial lobe is the most developed & gives rise to a prominent facial ridge.

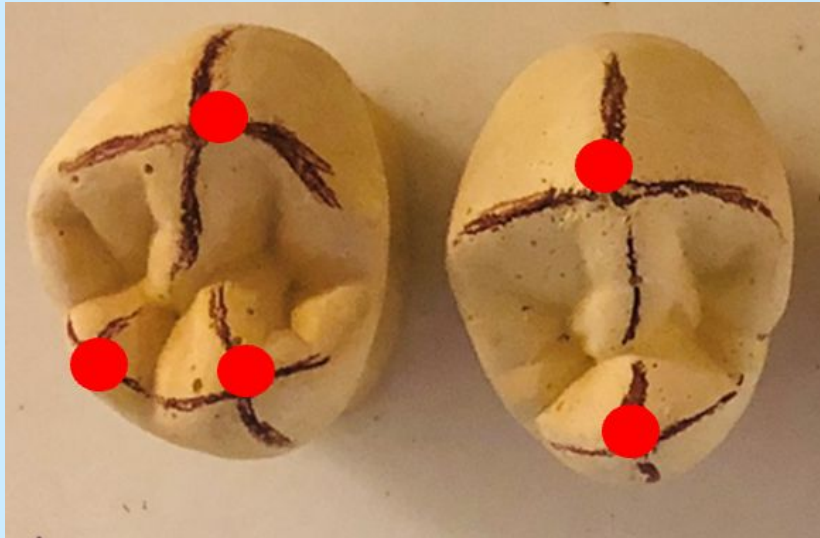


Developmental Lobes

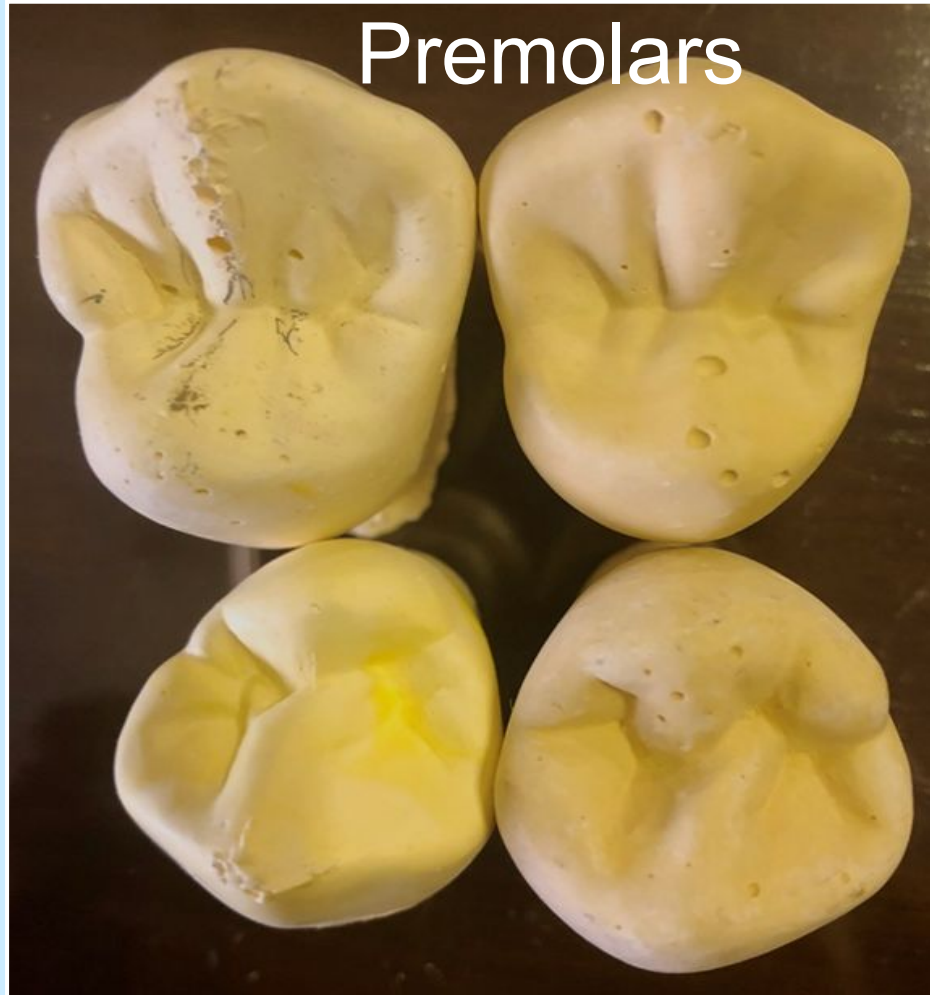
- Mandibular second premolars have 5 lobes in 3-cuspid type.



Class Traits & Key Considerations:



Term “Bicuspid” misleading – may have 2–3 cusps.



Class Traits & Key Considerations:

Maxillary premolars are morphologically more similar to each other than the mandibular premolars

Maxillary First Premolars

Class Traits & Key Considerations:

- It emerges into the oral cavity after the permanent maxillary lateral incisor but before the permanent maxillary canine.



DIMENSIONS:

Class Traits & Key Considerations:

**Cervico-incisal
length of crown**

8.5 mm

Length of root

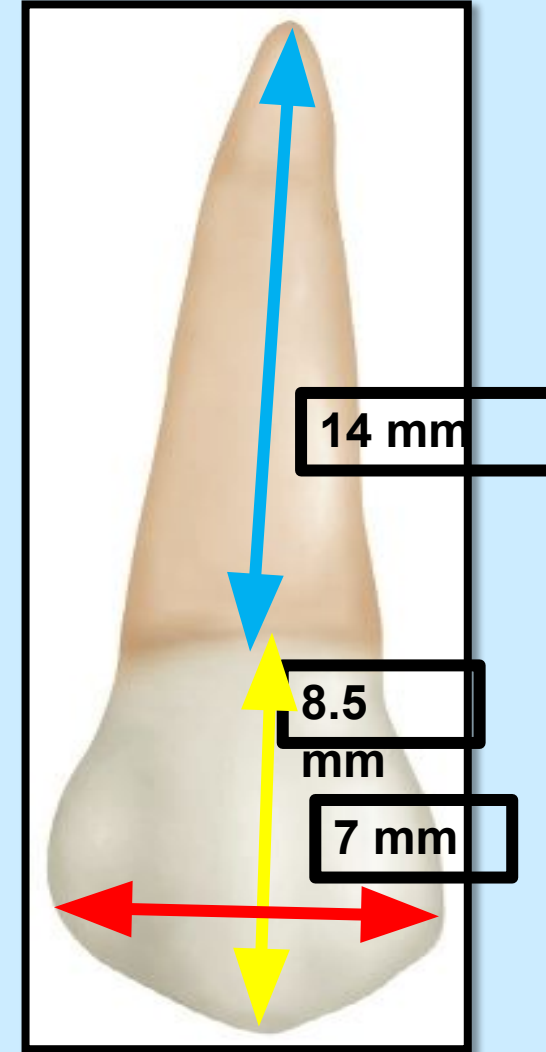
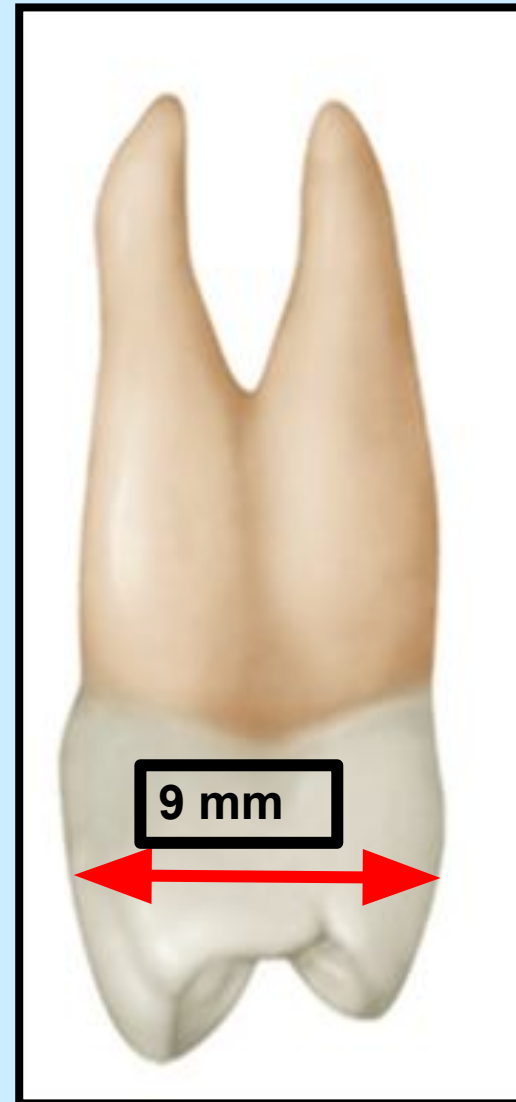
14 mm

**Mesiodistal
Diameter of Crown**

7 mm

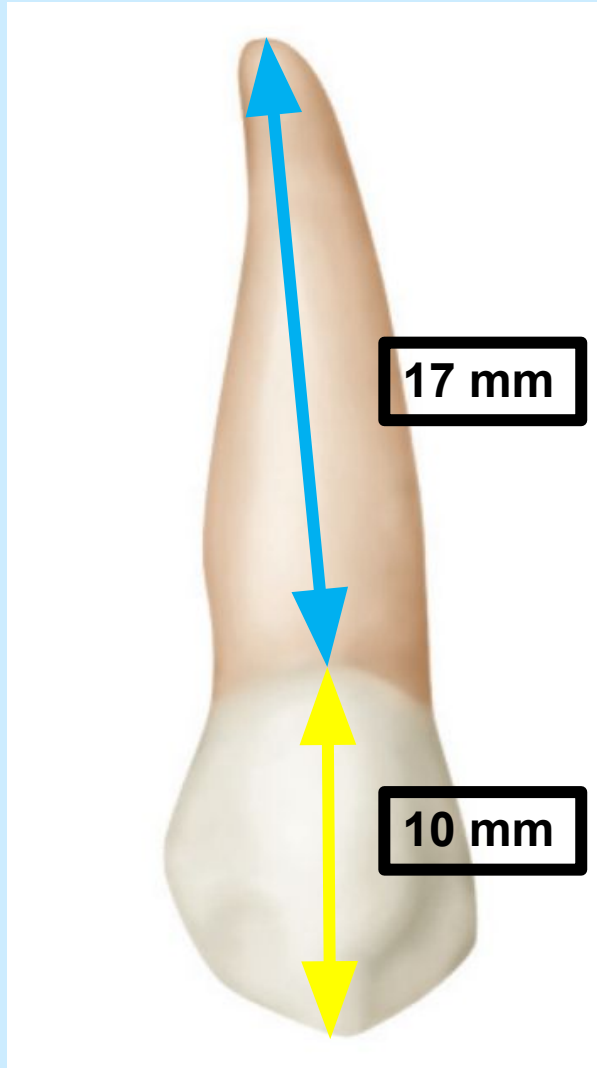
**Labiolingual
Diameter of crown**

9 mm



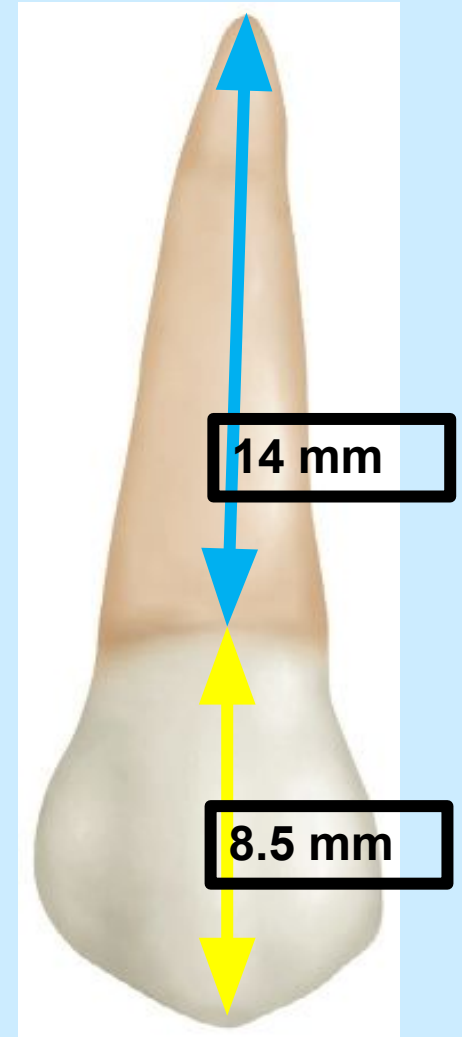
Measurement	Maxillary canine	maxillary 1 st premolar	Maxillary 1 st molar
Length of crown	10	8.5	7.5
Length of root	17	14	B=12,L=13
Mesiodistal diameter of crown	7.5	7	10
Mesiodistal diameter at cervix	5.5	5	8
Labiolingual diameter of crown	8	9	11
Labiolingual diameter at cervix	7	8	10
Cervical line curvature (mesial)	2.5	1	1
Cervical line curvature (distal)	1.5	0.0	0.0

COMPARISON WITH CANINES



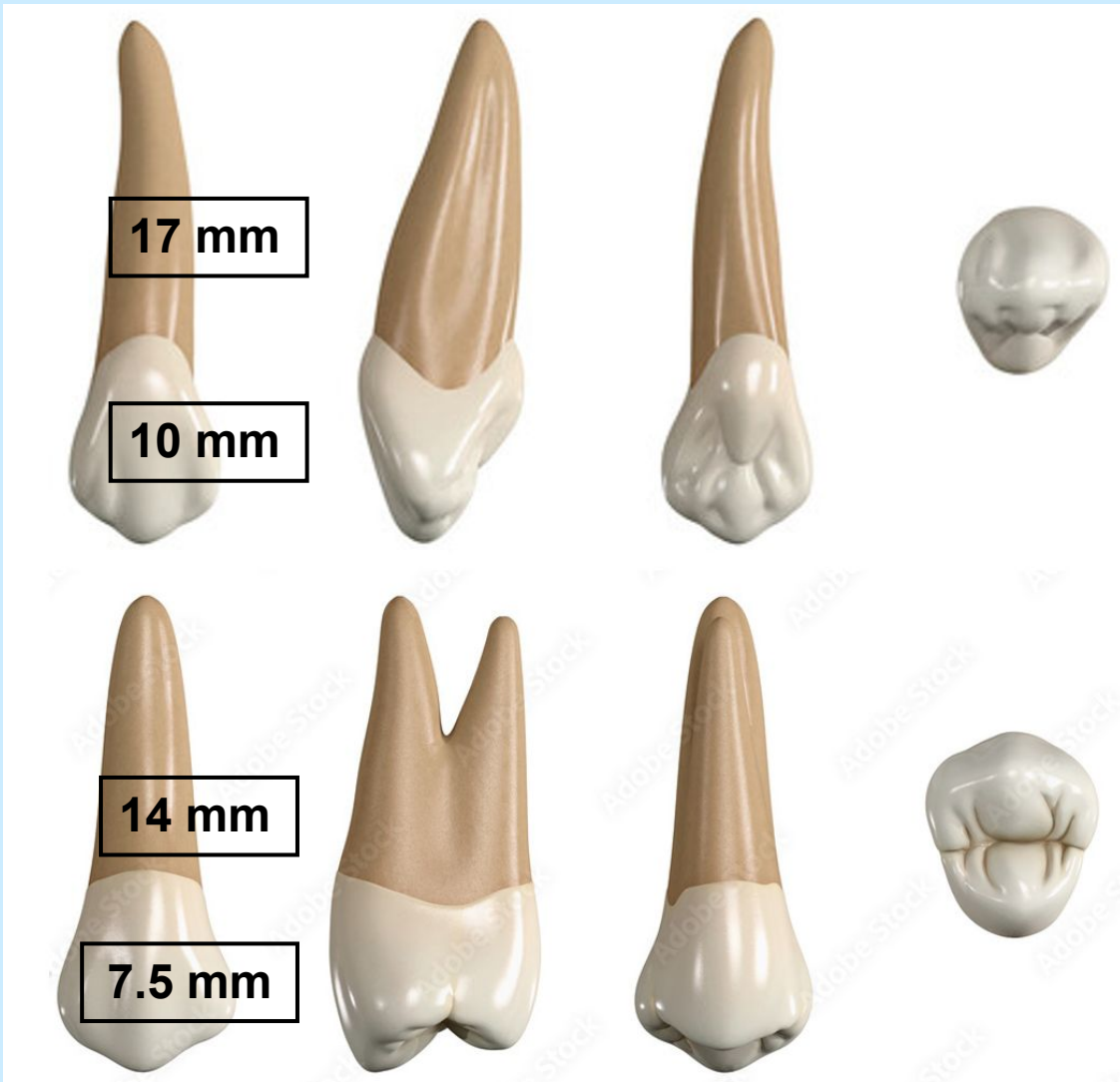
CANINE

Max.premolar
crown and
root length
shorter than
max.canine



Premolar

Comparison with max. Canine



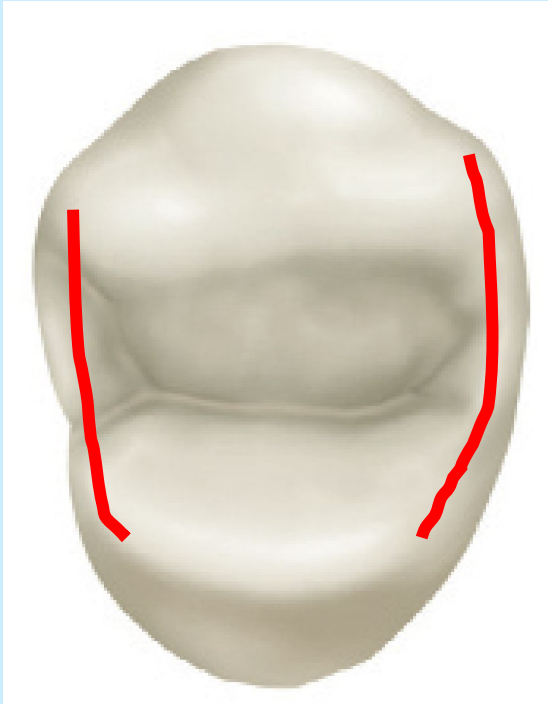
Premolar

Class Traits & Key Considerations:

- Shorter crown length (by 1.5 to 2 mm on the avg)
- Marginal ridges horizontal—part of occlusal surface.

Class Traits & Key Considerations:

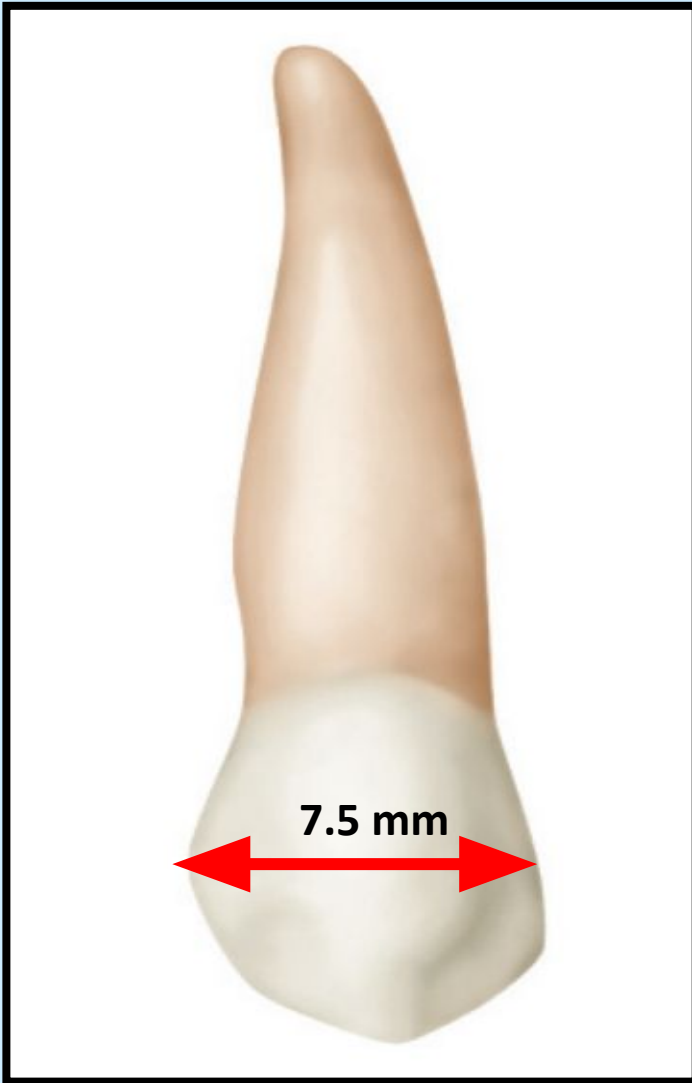
Marginal Ridges in Premolars:



Since cusps are well developed B & L,
So **marginal ridges** are horizontal & are considered part of the occlusal surface rather than being on lingual surface, as in incisors and canines.



Comparison with max. Canine



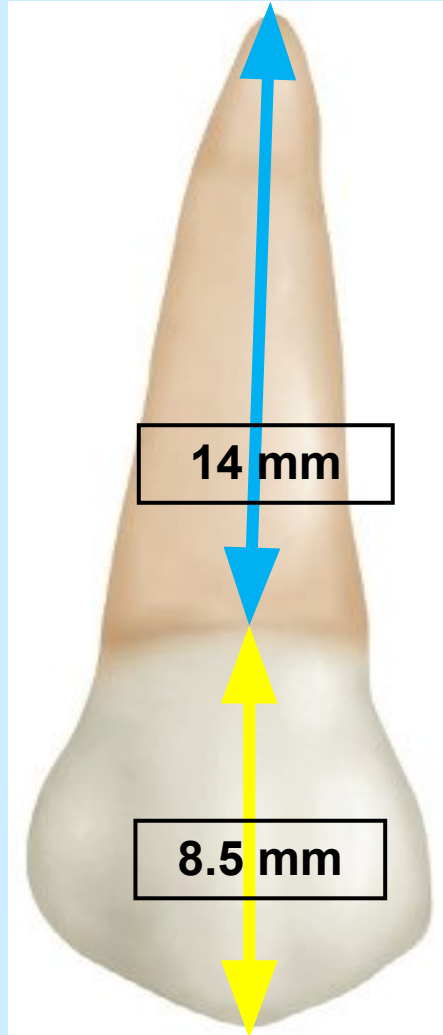
✓ Max. 1st PM not as wide in a MD direction as the canine.

Class Traits & Key Considerations:



Premolar

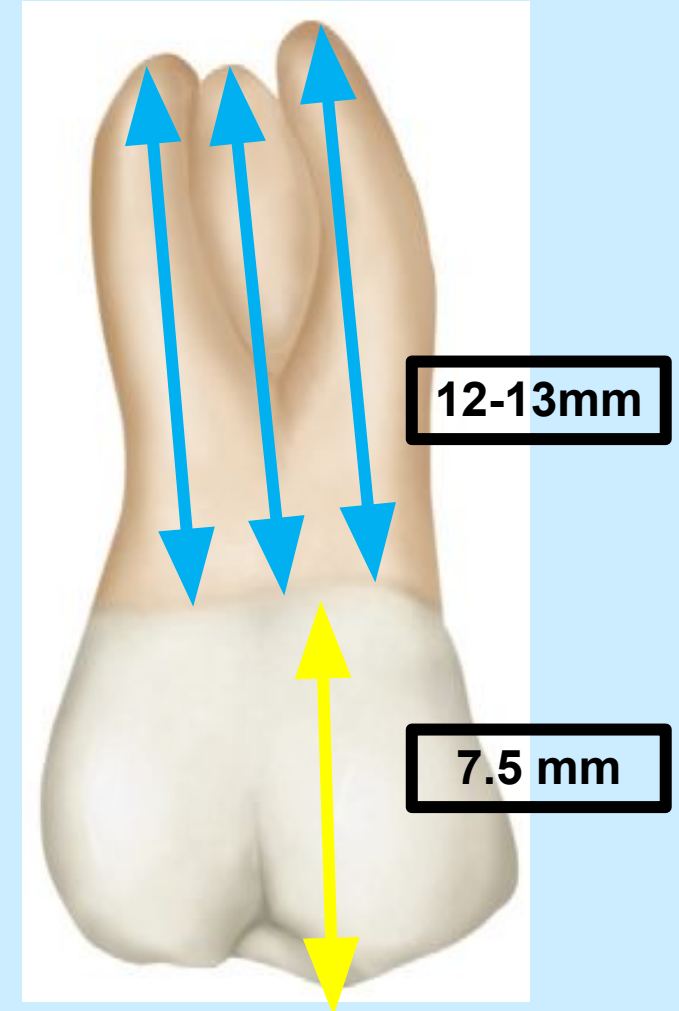
Comparison with Molars



Premolar

Class Traits & Key Considerations:

- Max.premolar crown and root length longer than molar.



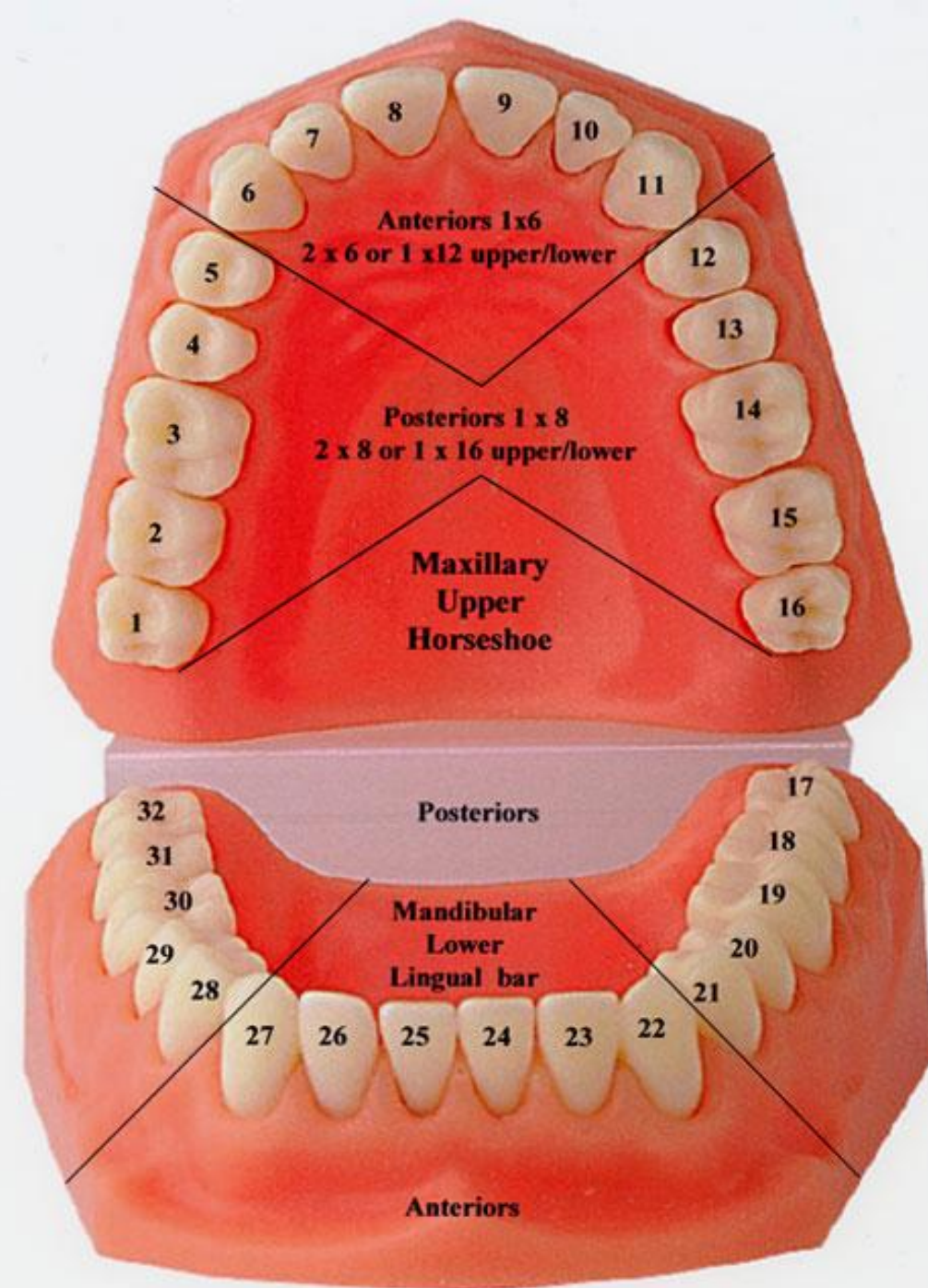
Molar

Morphology of Maxillary First Premolars



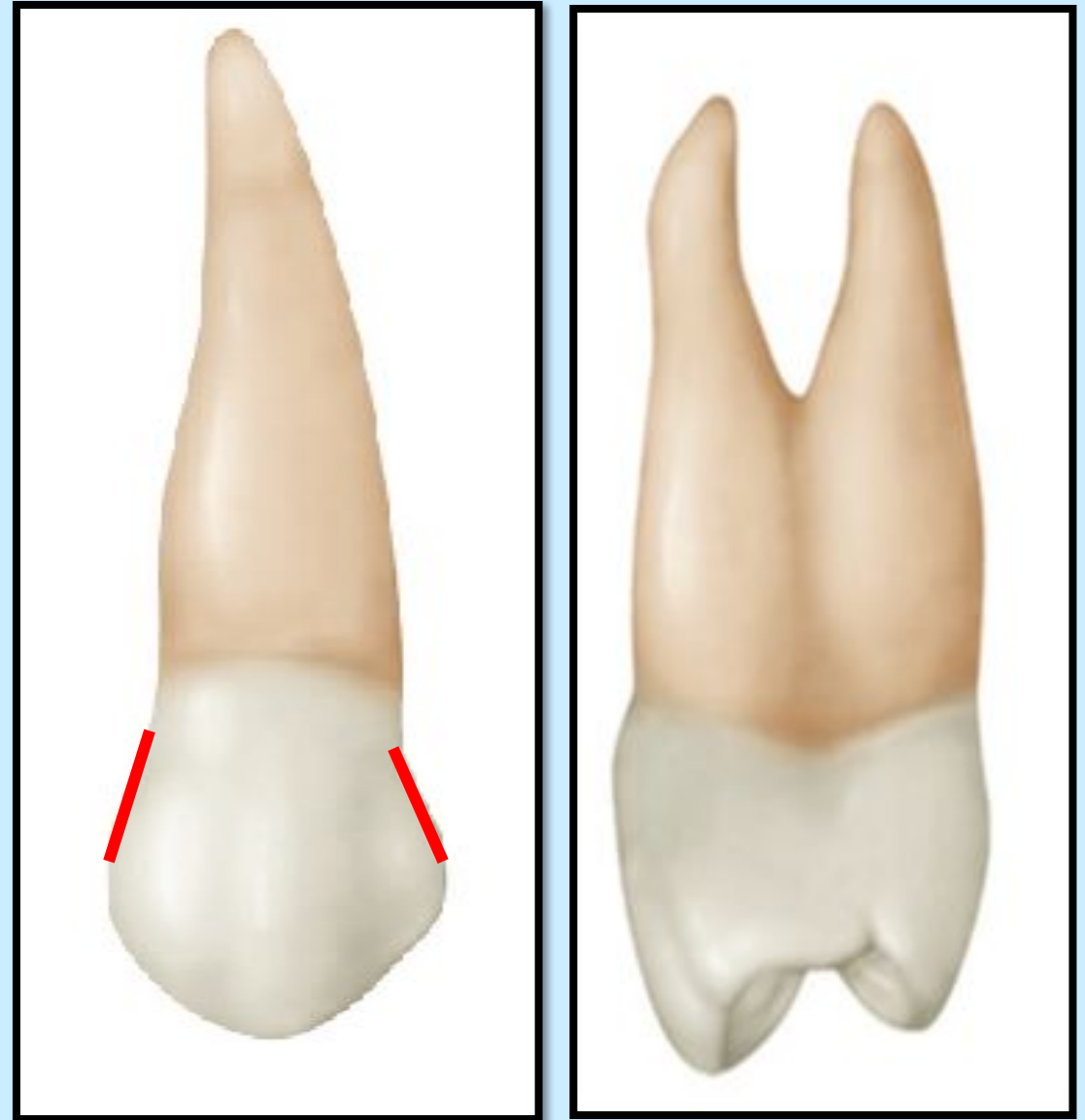
Maxillary First Premolars

- Tooth **#4** and **#13** (**Universal Numbering System**)
- The **first posterior** tooth in the maxillary arch.



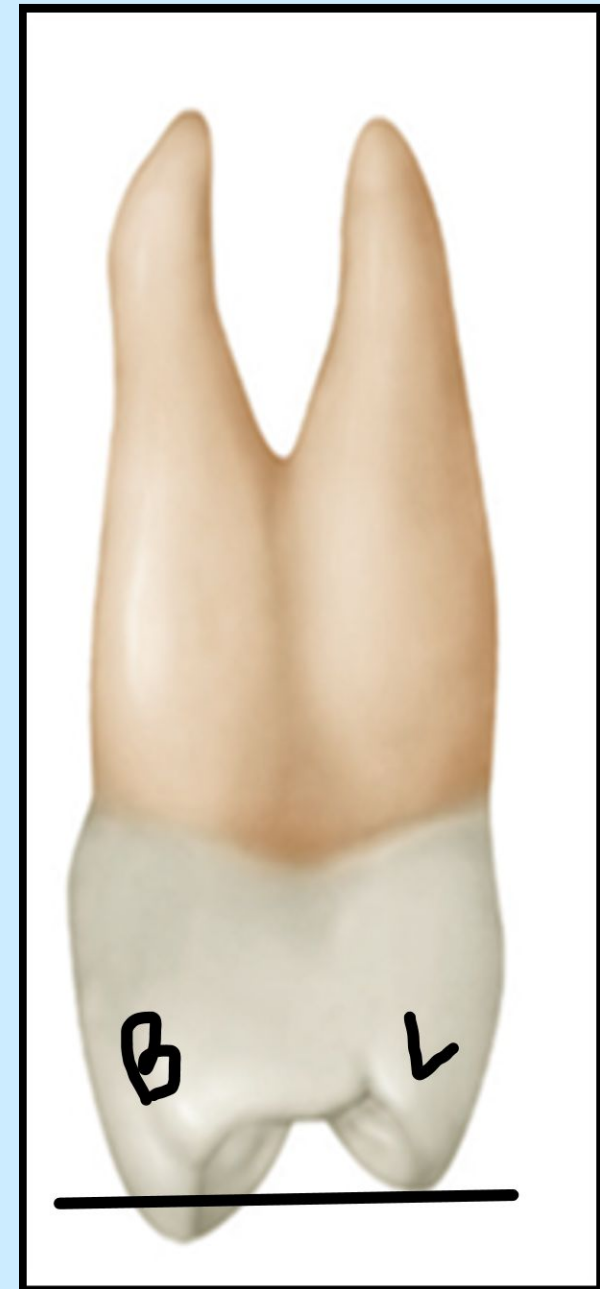
Crown Morphology

- Has two major cusps: a prominent **Buccal** and a shorter **Lingual**.
- Crown appearance is **angular** with **prominent buccal line angles**.



Cusp Features:

- Buccal cusp is **~1 mm longer** than the lingual cusp.

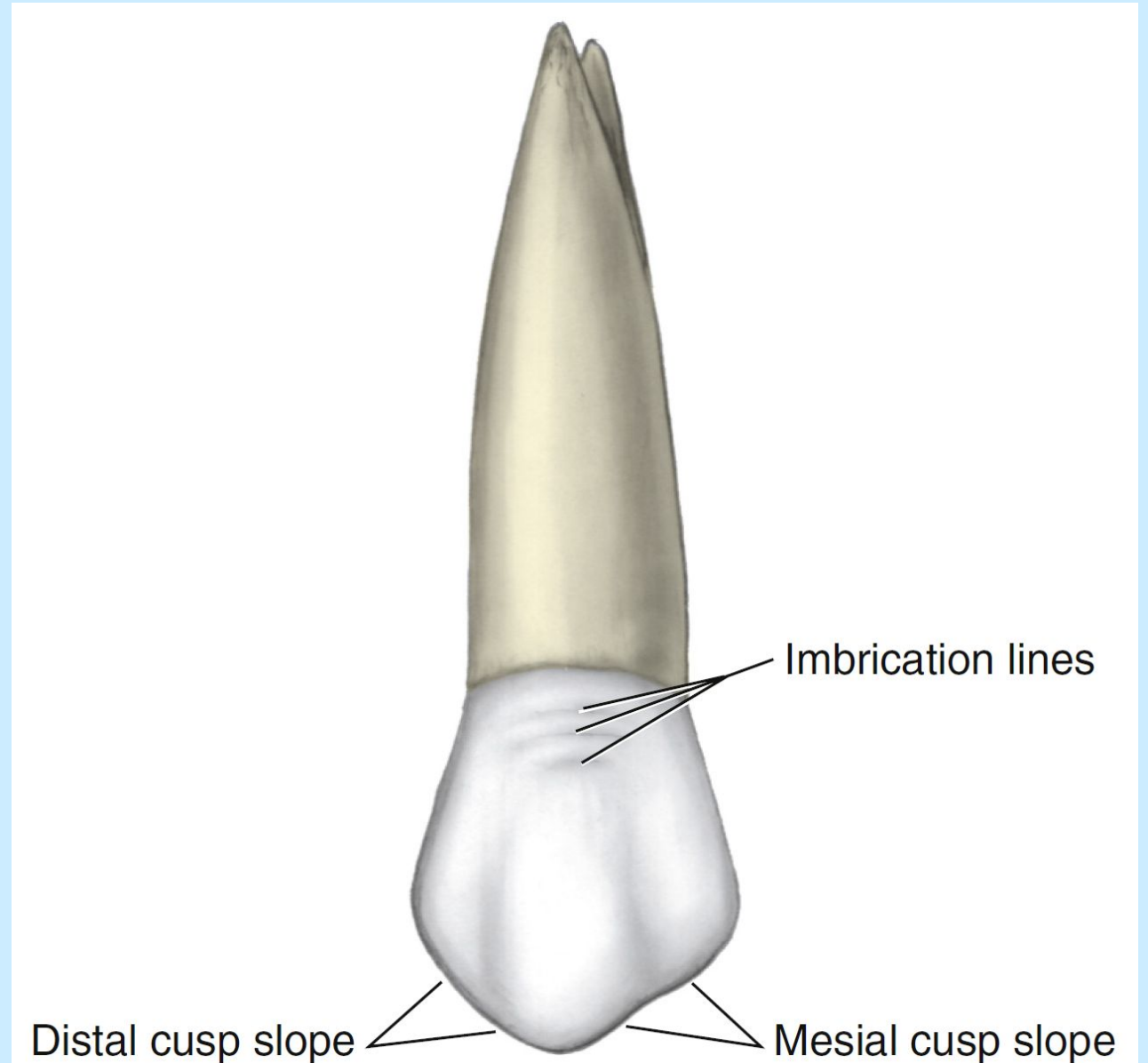


Cusp Features:

Mesial slope of the buccal cusp is **longer** than the distal slope.

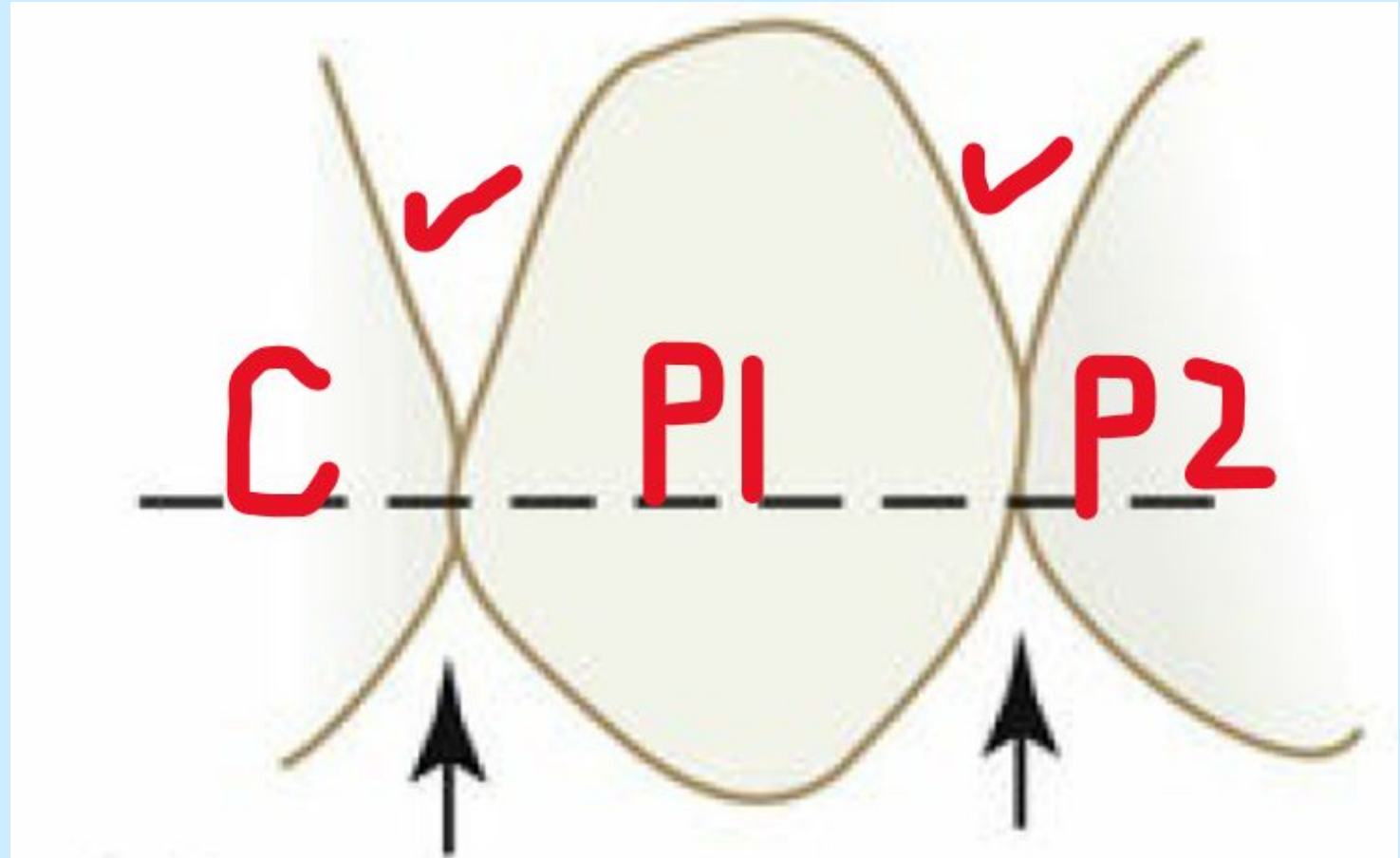
(This is the opposite of the maxillary canine)

FIGURE 17-13 Buccal features of the permanent maxillary right first premolar.



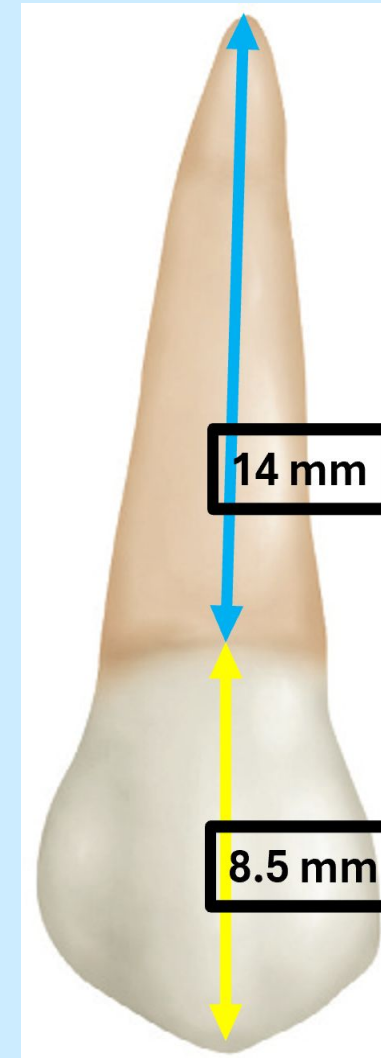
Proximal Contacts:

Mesial and distal contact areas are at **approximately the same level** cervico-occlusally .

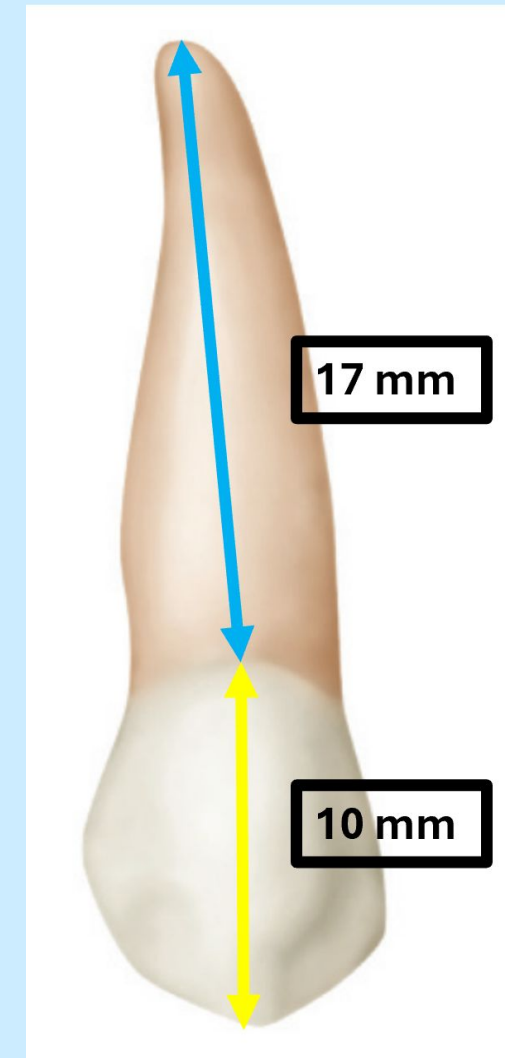


Differentiating from Anterior Teeth (Posterior Tooth Traits)

- ✓ Crown is **shorter** than the maxillary canine by **1.5 - 2 mm** on average.
- ✓ Buccal root **3-4 mm short** of maxillary canine root length.
- ✓ **Narrower** mesiodistally than the canine.



Premolar



Canine

Differentiating from Anterior Teeth (Posterior Tooth Traits)



Canine

**Diamond
shape of
occlusal table**



Premolar

**hexagonal shape
of occlusal table**

Differentiating from Anterior Teeth (Posterior Tooth Traits)



Canine

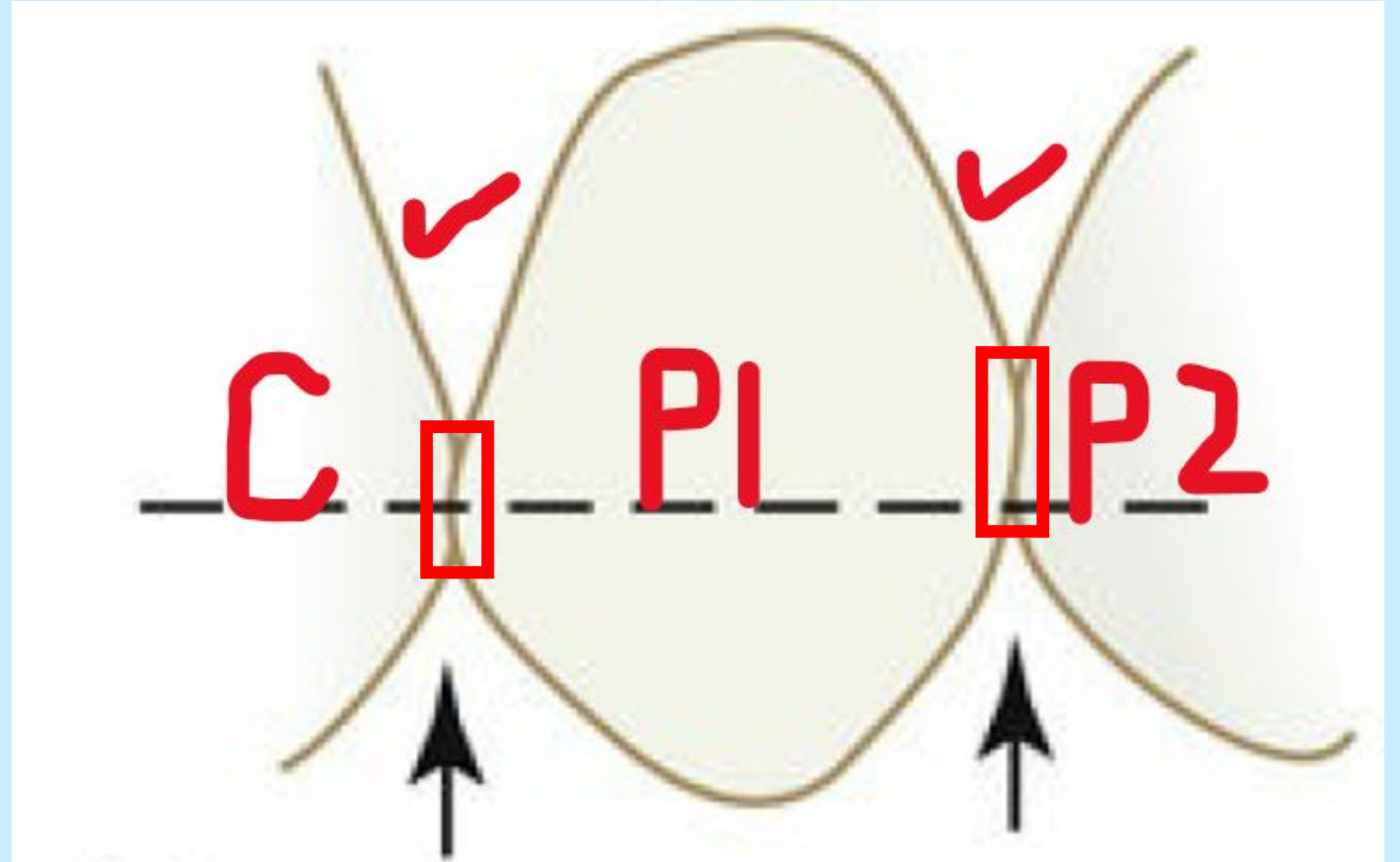


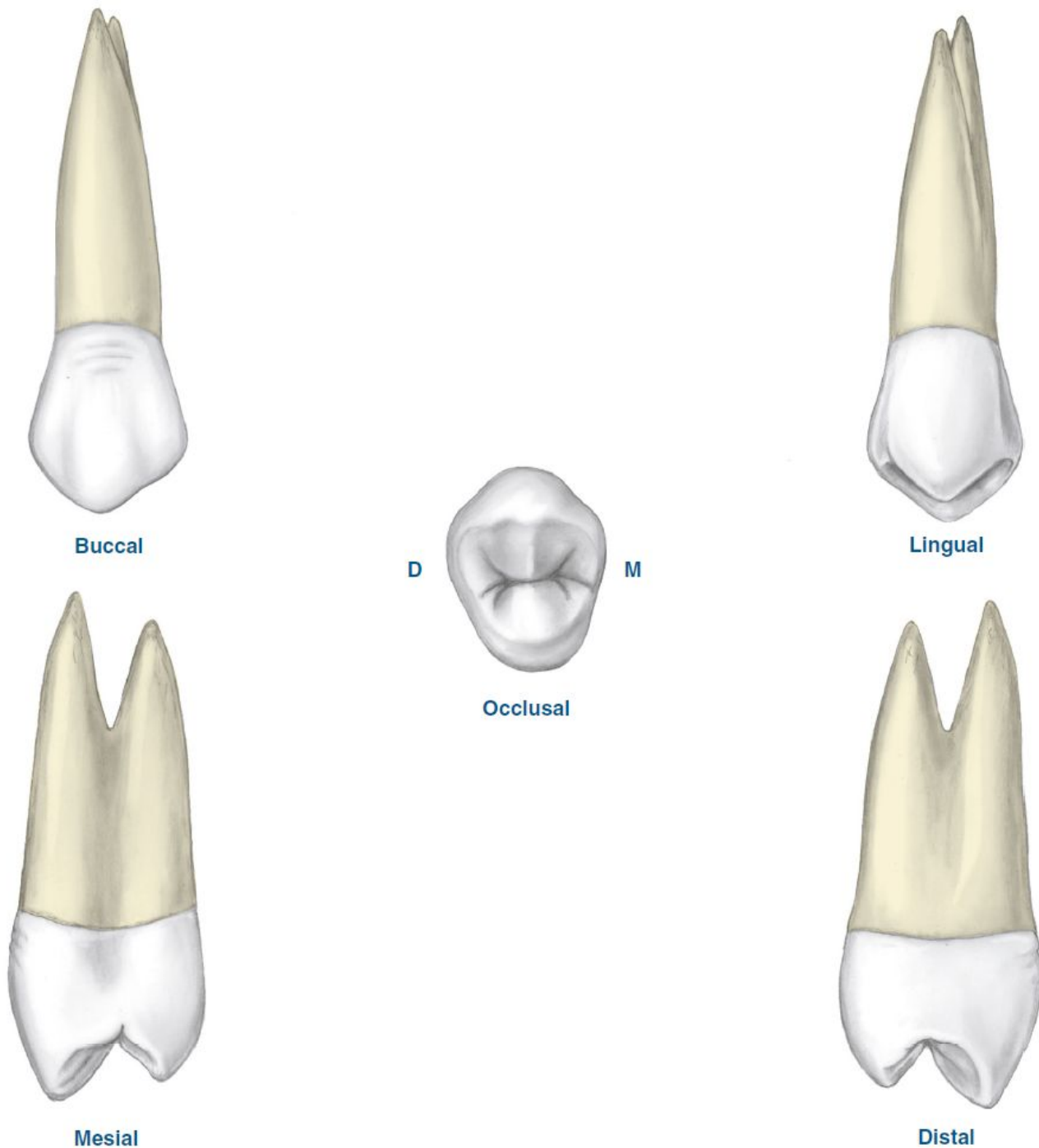
Premolar



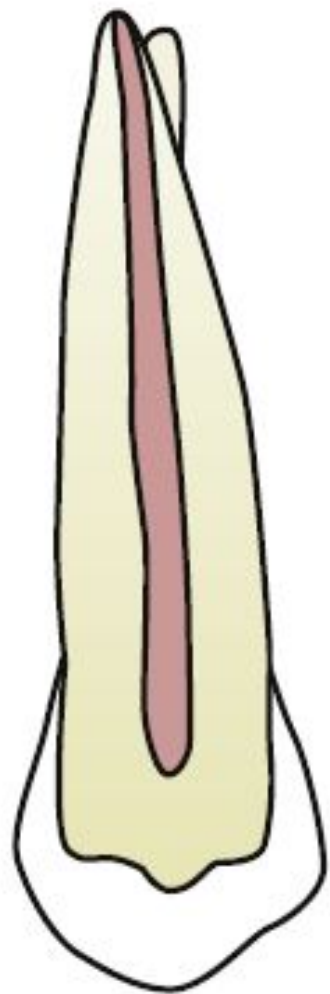
Presence of **well-defined occlusal table** with cusps.

Broader contact areas compared to anterior teeth.





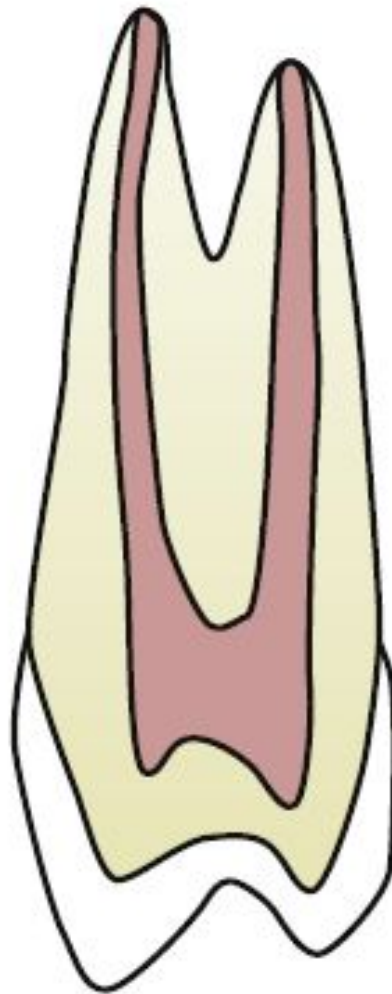
•**FIGURE 17-11** Views of the permanent maxillary right first premolar.



**Mesiodistal
section**

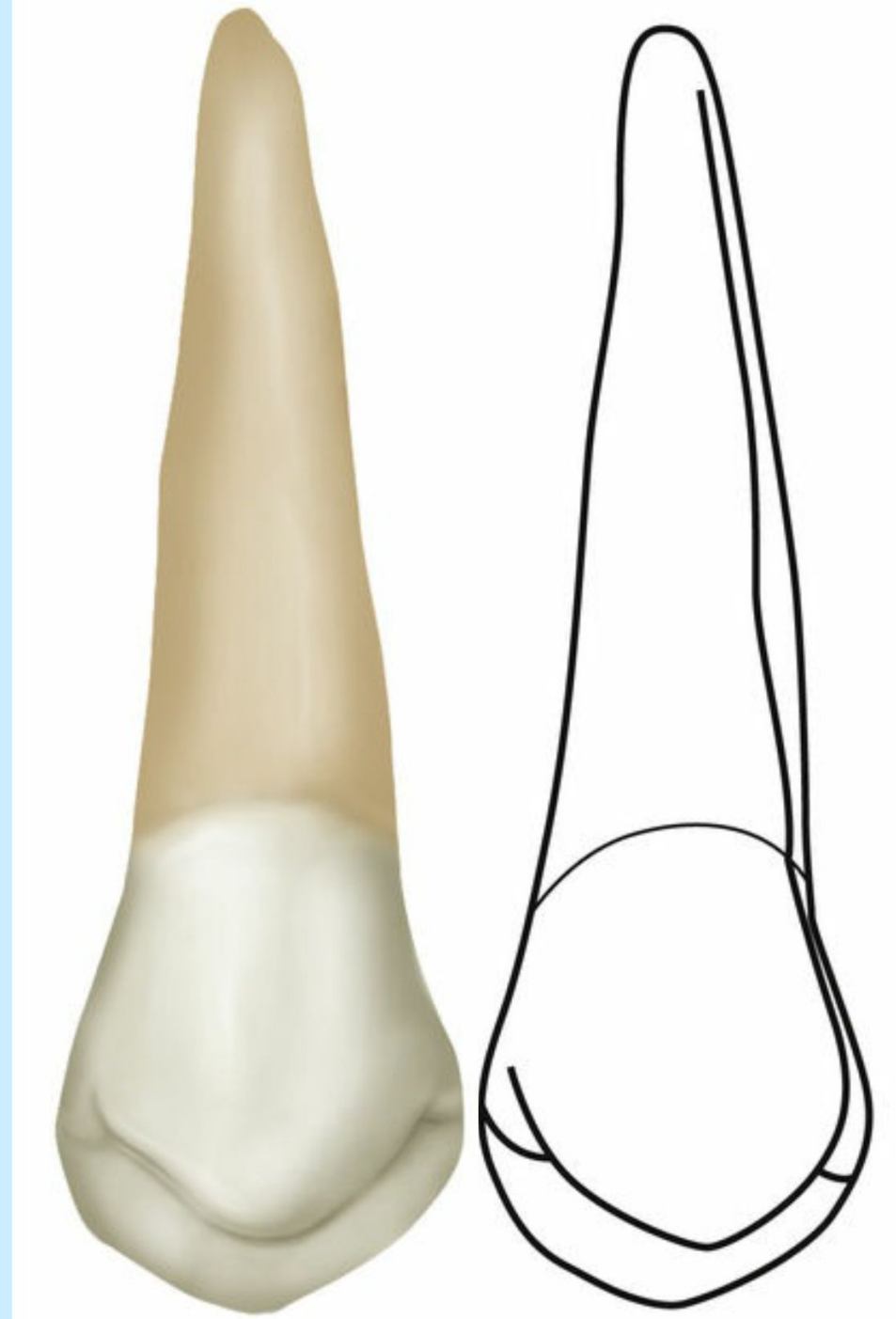
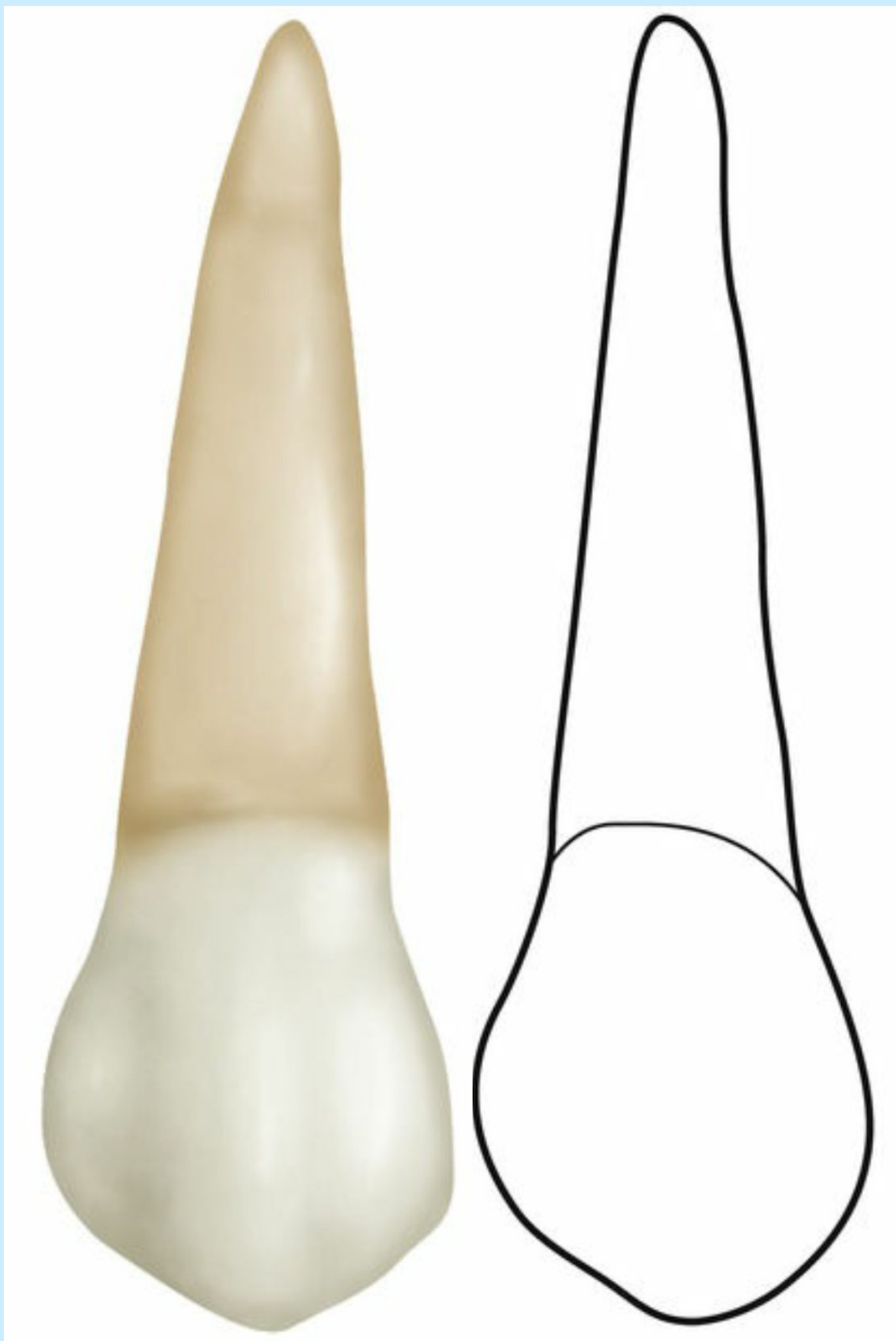


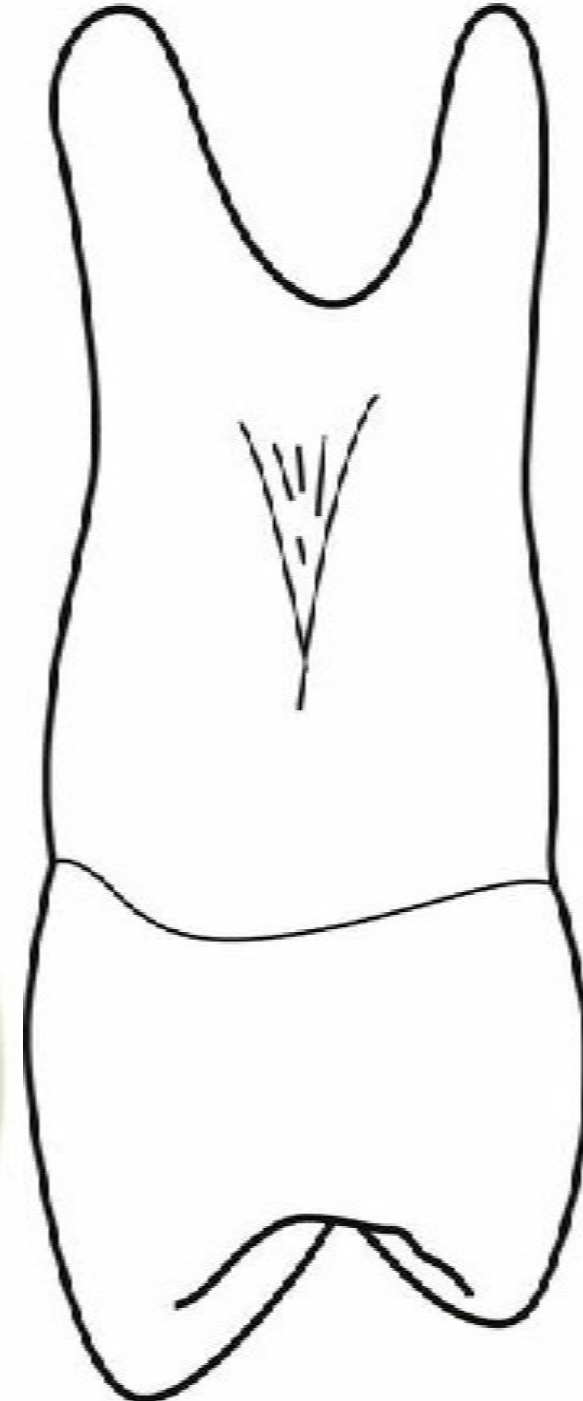
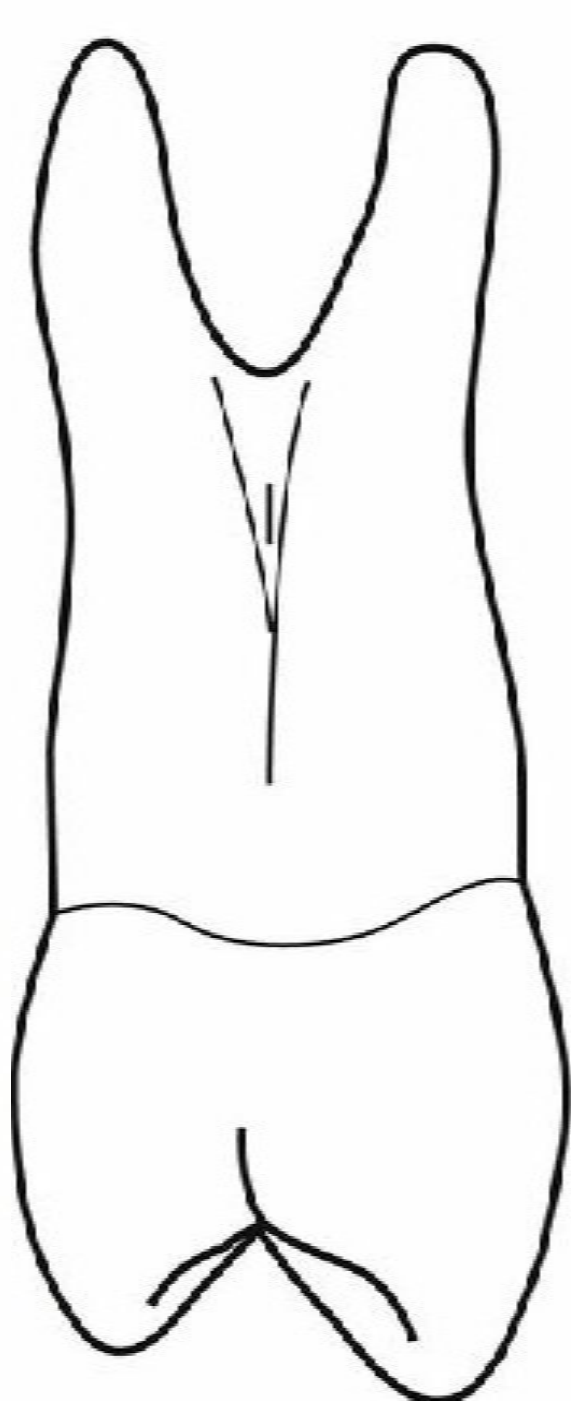
**Cervical
cross section**

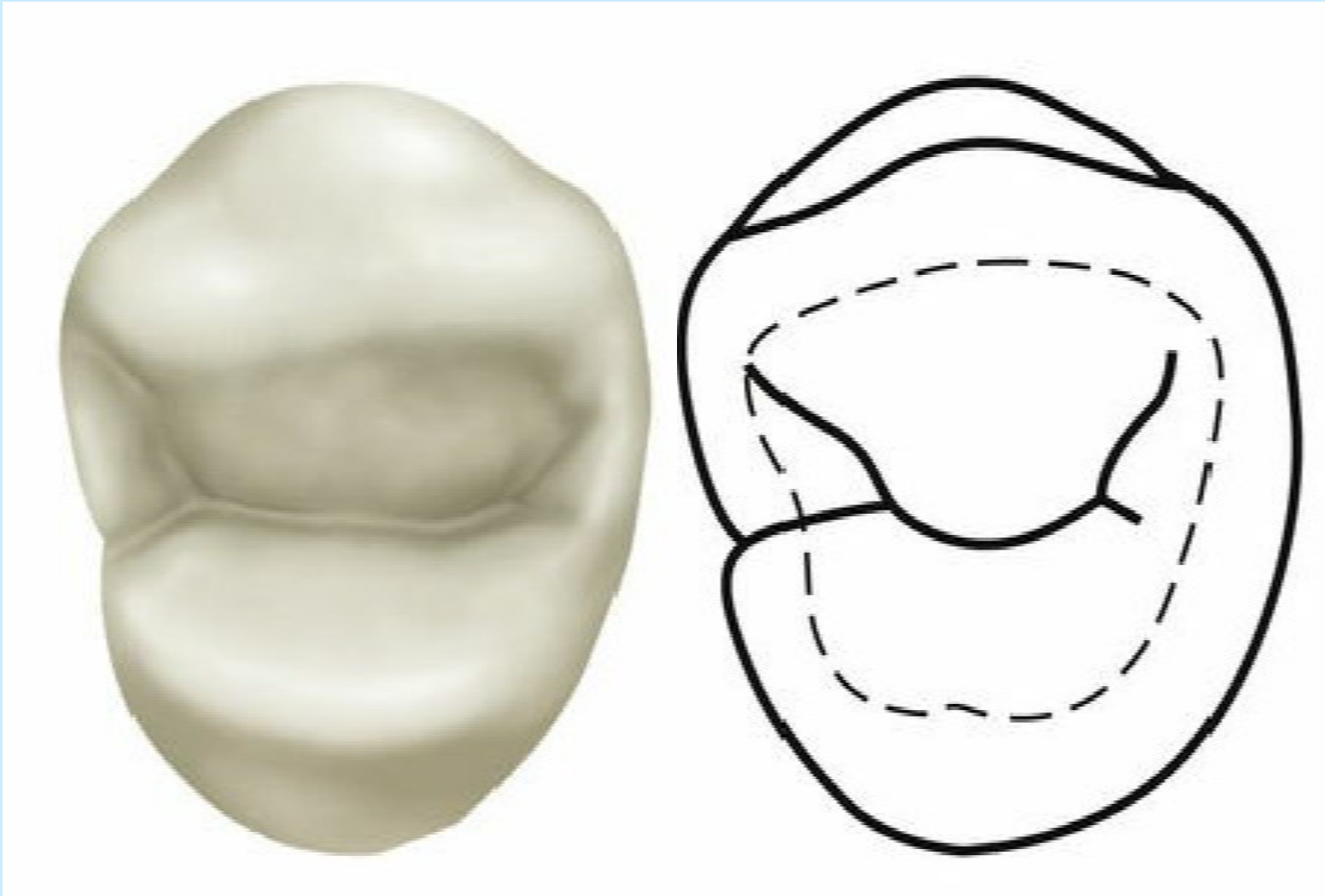


**Buccolingual
section**

FIGURE 17-12 Pulp cavity of the permanent maxillary right first premolar.

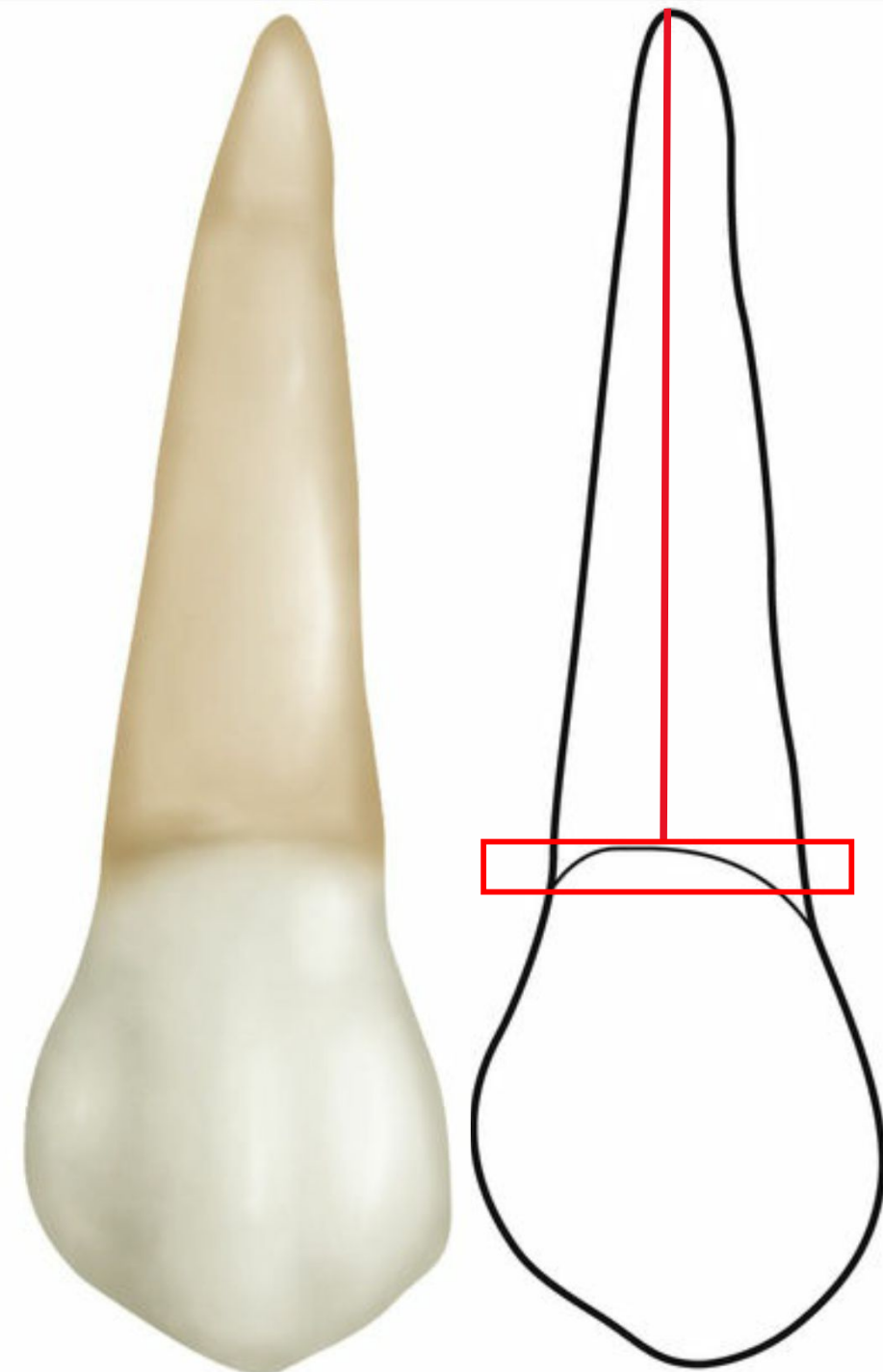
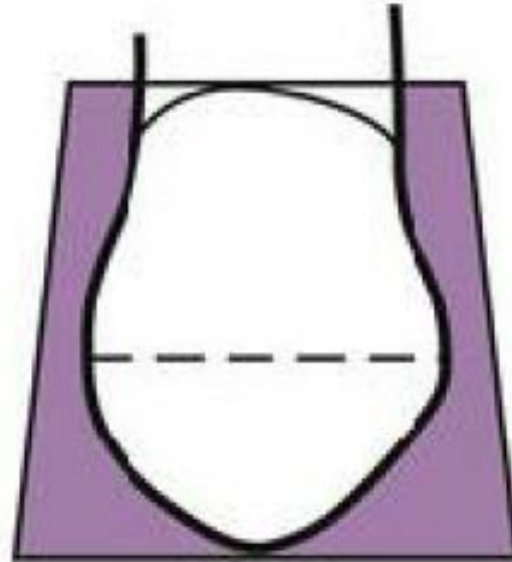






Overall Crown Shape from the Buccal View

- **Shape:** Roughly **Trapezoidal** (without considering cuspal tips) or **Pentagon** (considering cuspal tips).
- **Cervical Line:** Exhibits little curvature with crest near center of root buccally.



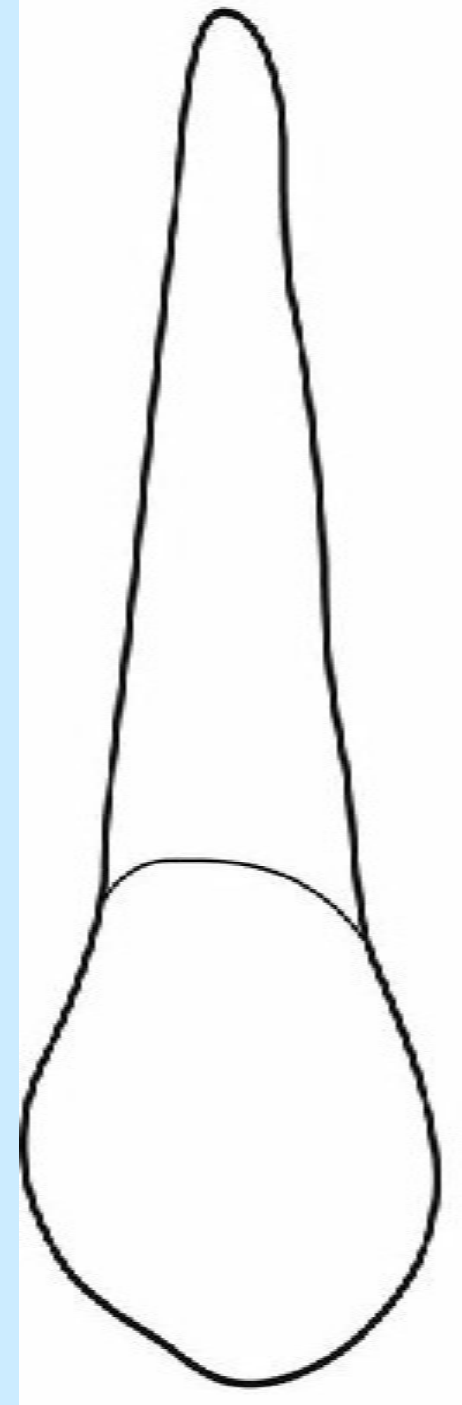
Buccal View

- **Buccal Ridge:** A prominent, convex ridge runs from the cusp tip to the cervical margin, representing the strong middle buccal lobe.
- **Developmental Depressions:** Found mesial and distal to the buccal ridge, demarcating the mesiobuccal and distobuccal lobes.
- **Line Angles:** Strong and prominent mesiobuccal and distobuccal line angles.



Buccal View

Feature	Mesial Outline	Distal Outline
Crown Contour	Slightly concave from cervix to contact area.	Straighter, but may be slightly concave.
Contact Area	Crest lies immediately occlusal to the halfway point. Broad contact area.	Broader curvature/ contact area than mesial; crest tends to be slightly more occlusal.
Cusp Slope	Longer and rather straight. May be notched or concave.	Shorter and more curved.

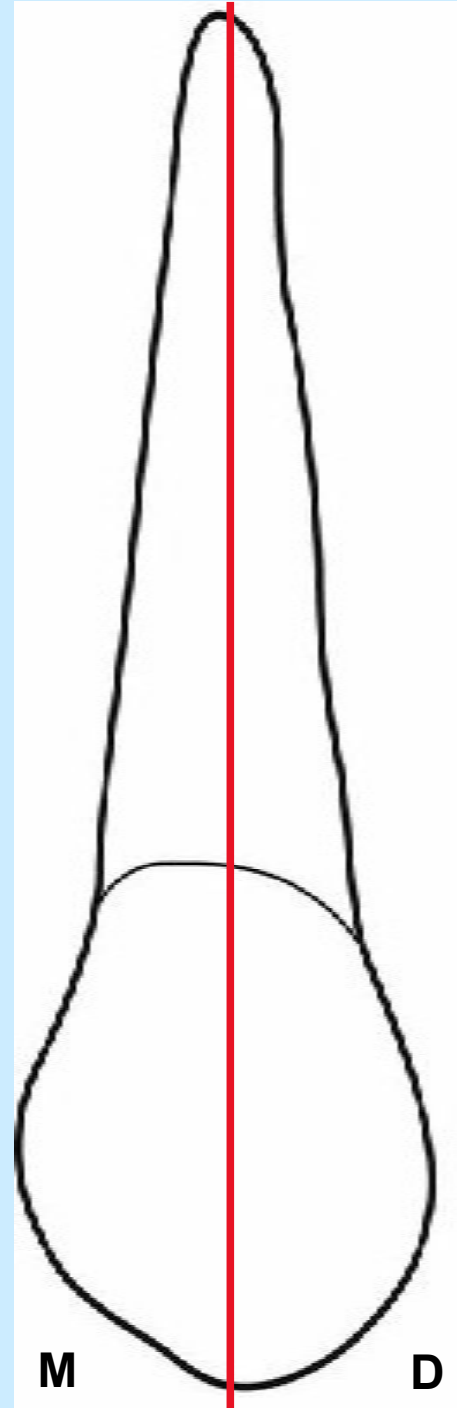


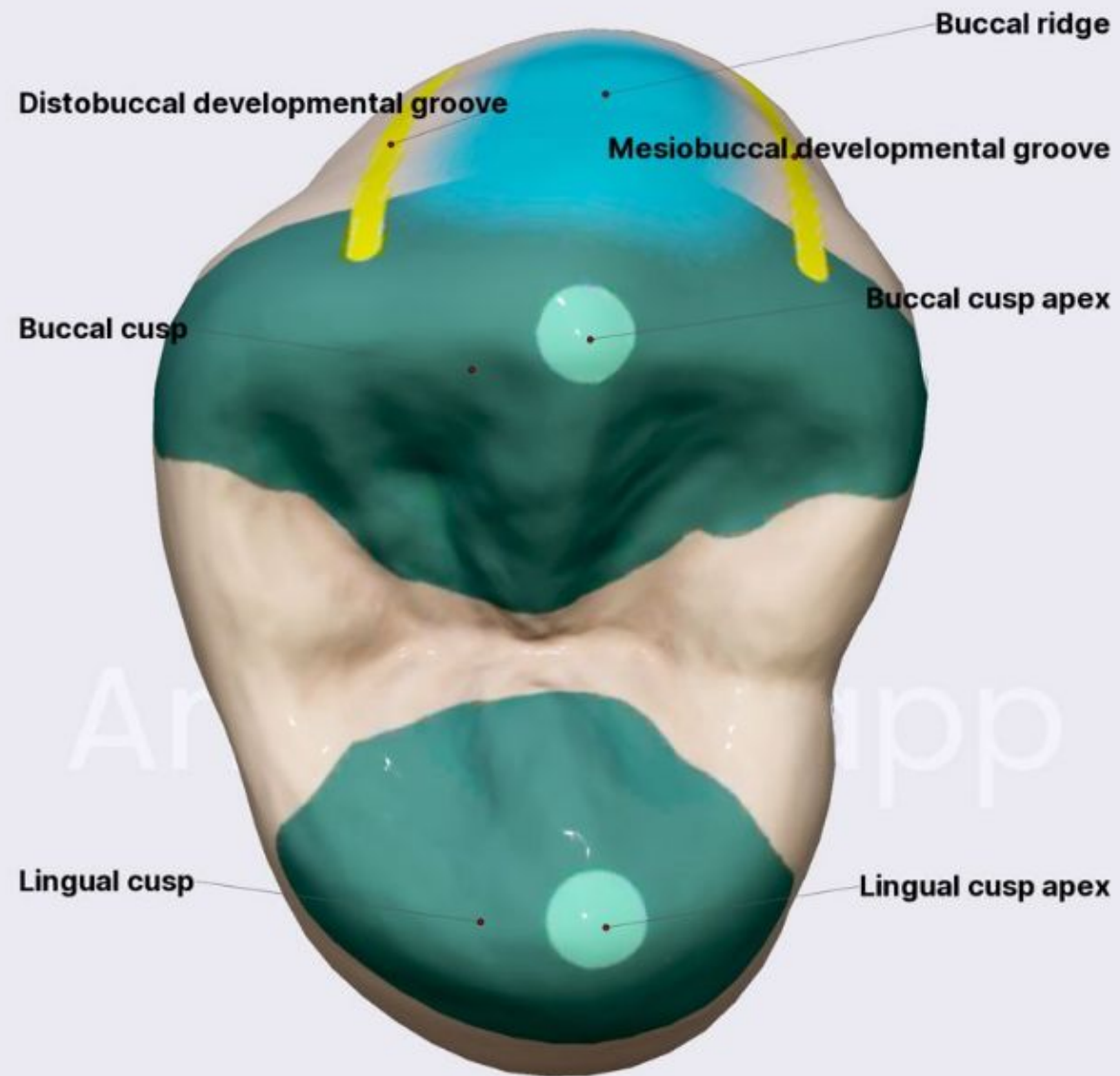
BUCCAL ASPECT

Buccal cusp Tip:

distal to a line bisecting the buccal surface of the crown.

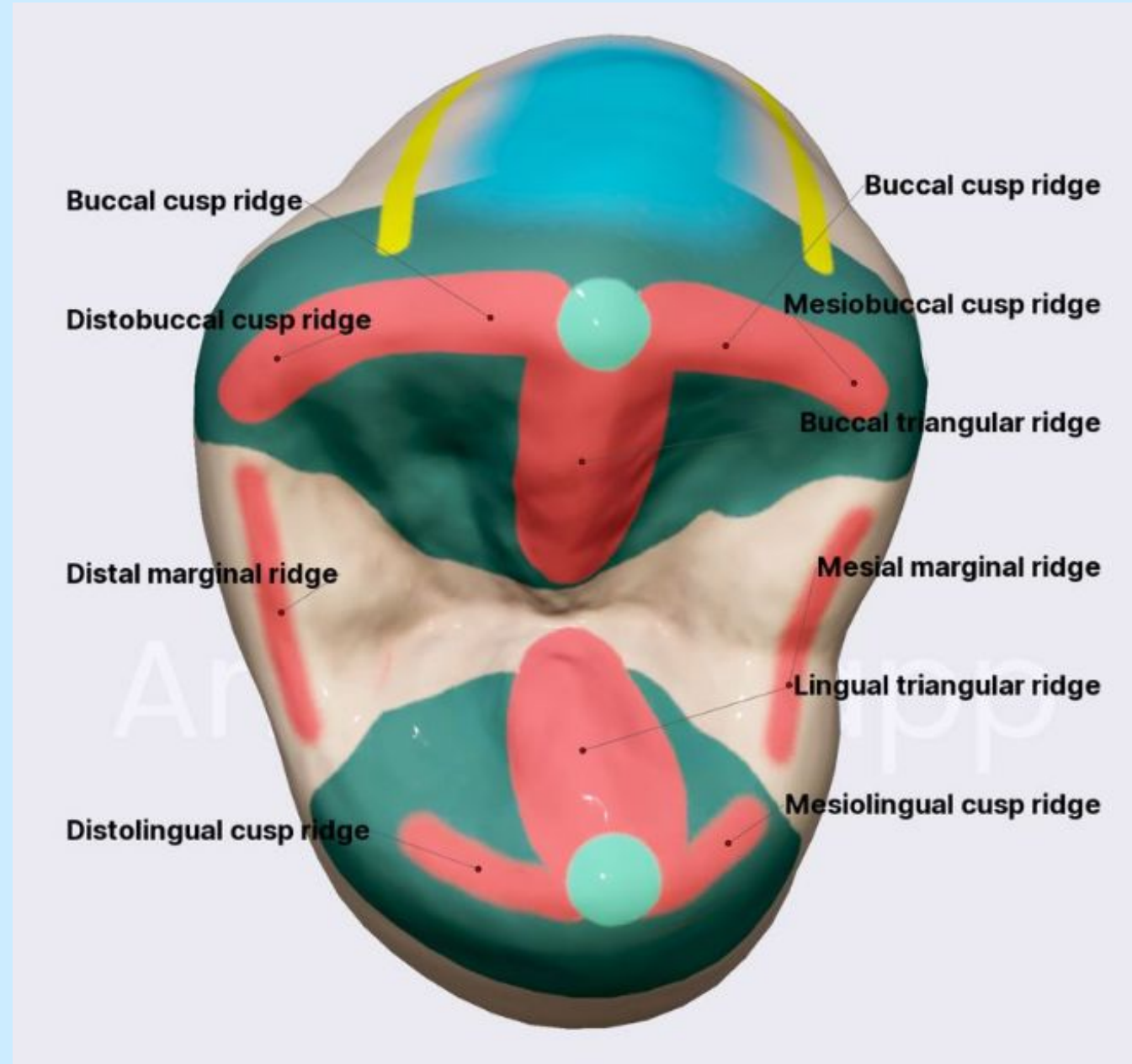
- **Mesial Cuspal Slope:** straight & long
- **Distal Cuspal Slope:** curved & short.





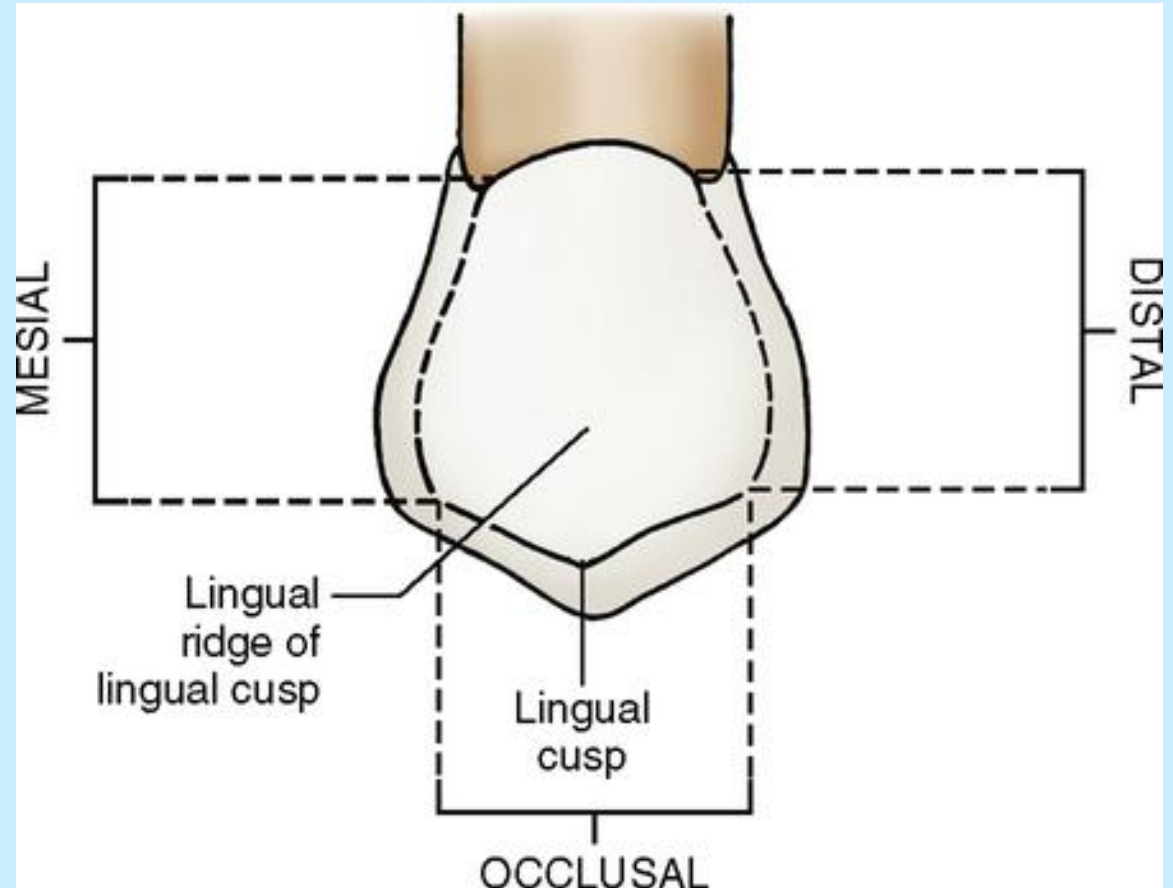
Buccal View (Summary)

- Buccal Cusp Tip
- Longer Mesial Slope
- Shorter, Curved Distal Slope
- Buccal Ridge
- Mesiobuccal Developmental Depression
- Mesial Contact Area
- Distal Contact Area
- Cervical Line



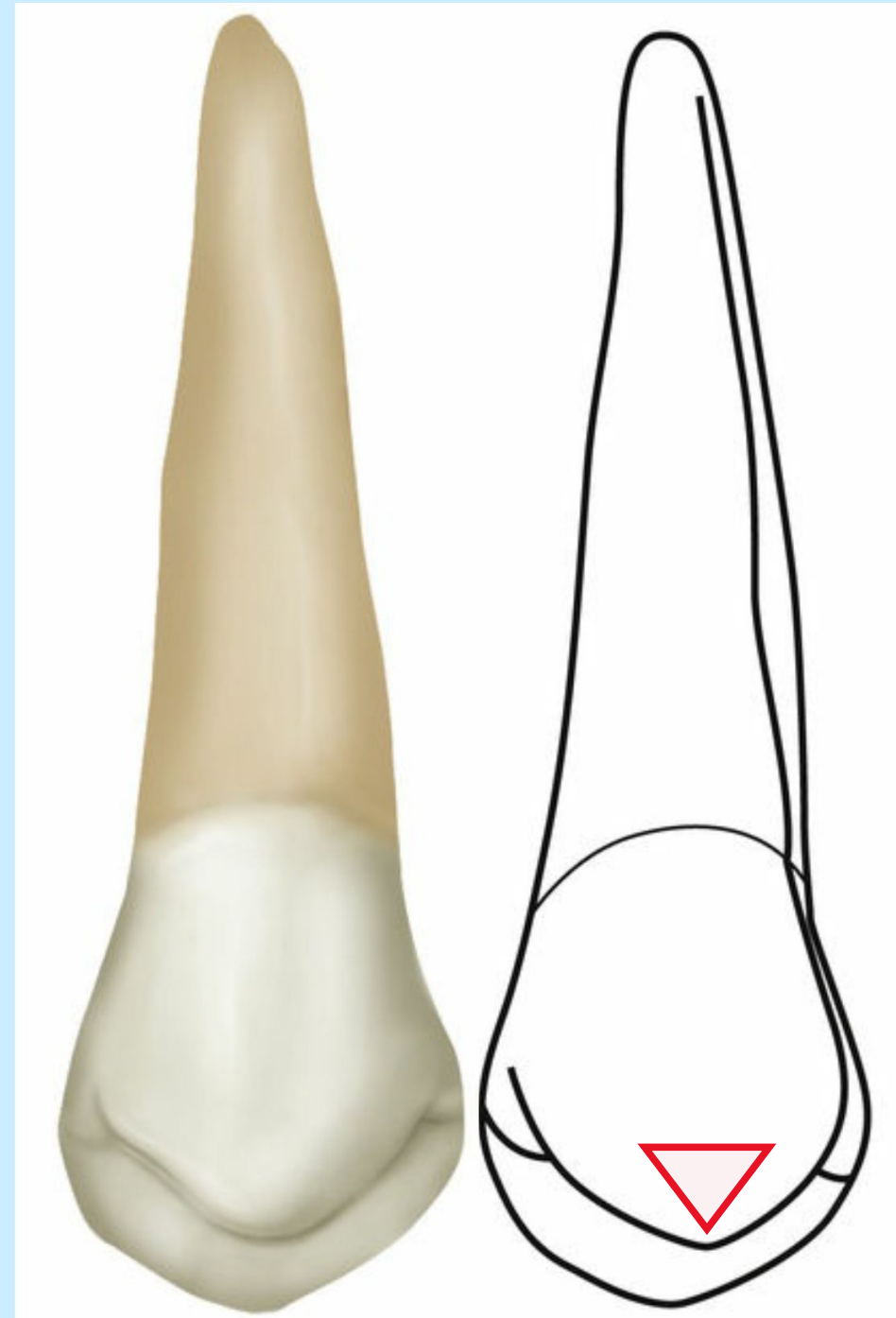
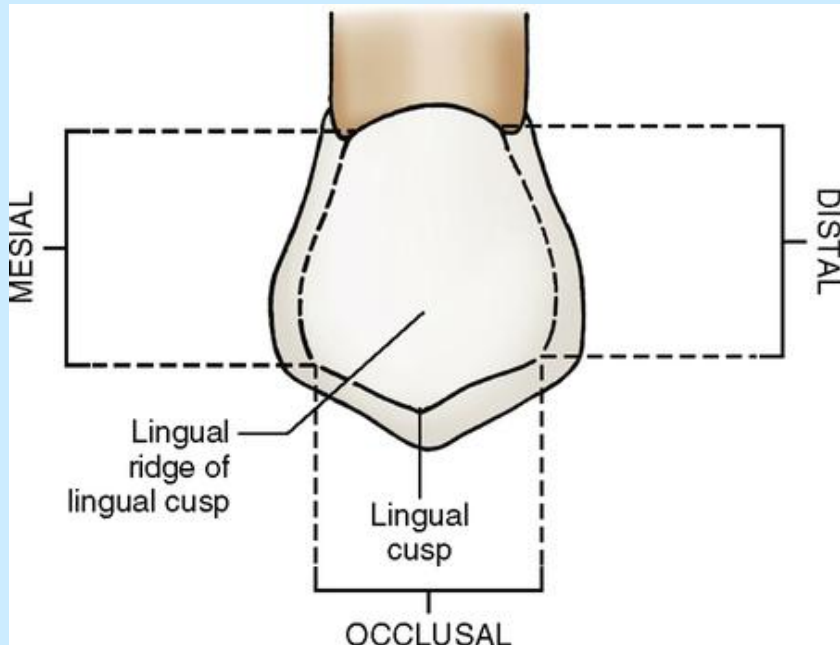
Lingual Aspect

- **Shape:** Reverse outline of the buccal aspect.
- **Size:** Shorter and narrower than the buccal cusp.
- **Form:** Smooth and **spheroidal** (rounded) from the cervix to near the tip.



Lingual Aspect

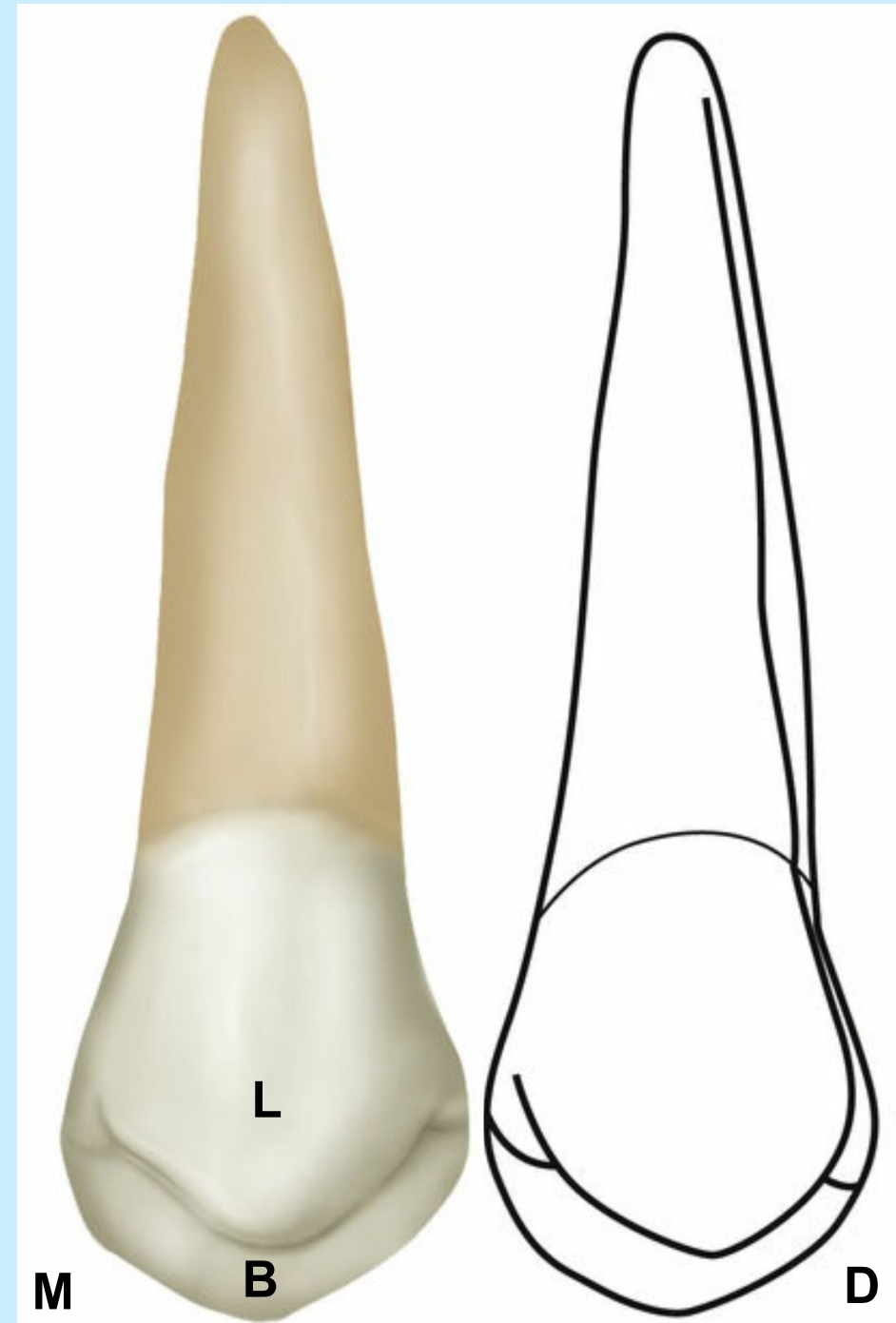
Sharp L cusp, M & D cuspal slope meeting at **90 degrees**



Lingual Aspect

Key Feature:

- Pronounced **taper** towards the lingual.
- The crown is significantly **narrower mesiodistally** on the lingual side than on the buccal side.

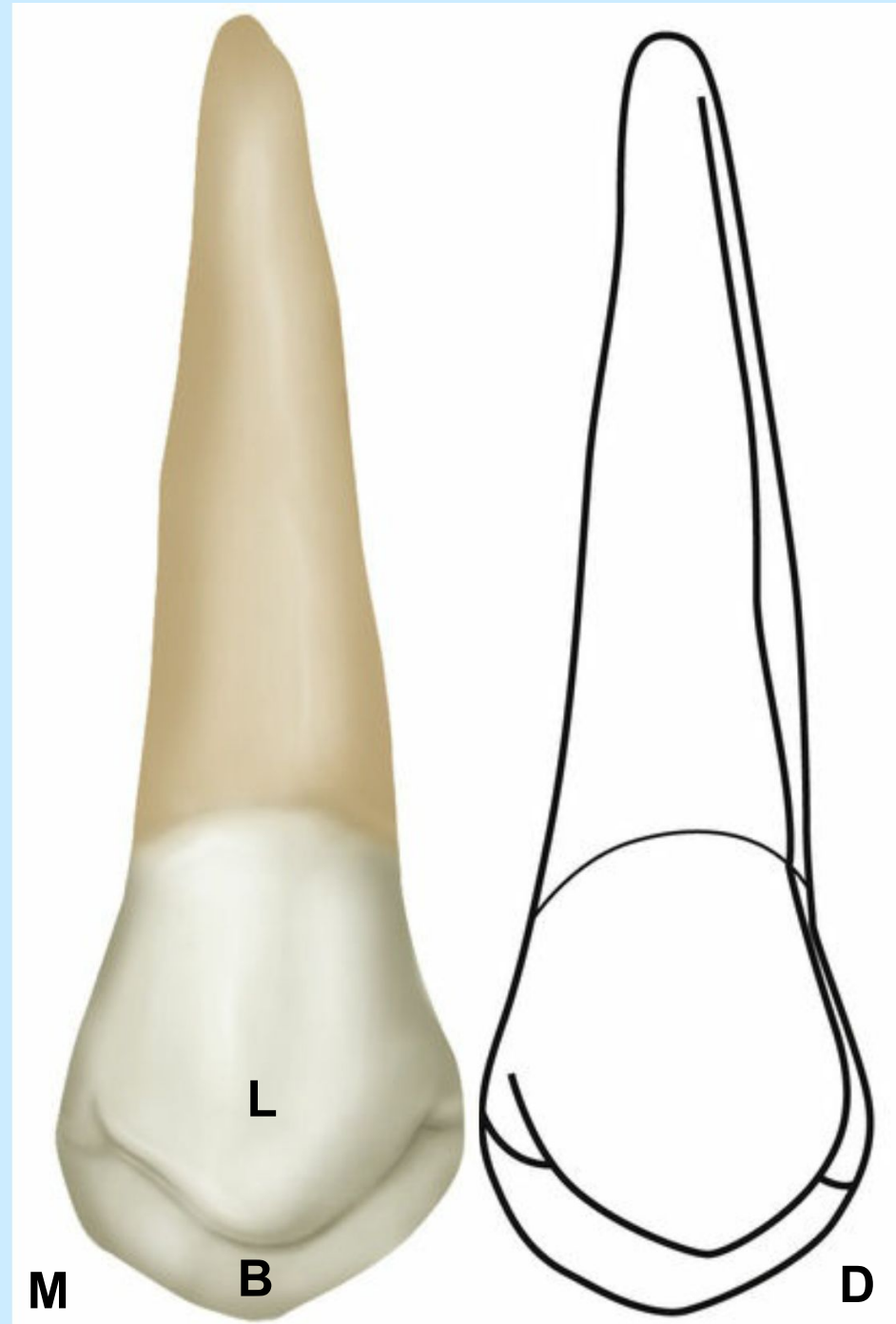


Lingual aspect

Lingual taper:

MD width of L cusp < MD of B cusp
1 mm shorter L cusp than B cusp.

Due to this taper, parts of the **mesial**
and distal surfaces of the crown
and root are visible from this aspect.



MESIAL ASPECT



BUCCAL ROOT

**MESIAL
DEVELOPMENTAL
DEPRESSION**

**BUCCAL
CERVICAL RIDGE**

**MESIAL CONTACT
AREA**

BUCCAL CUSP

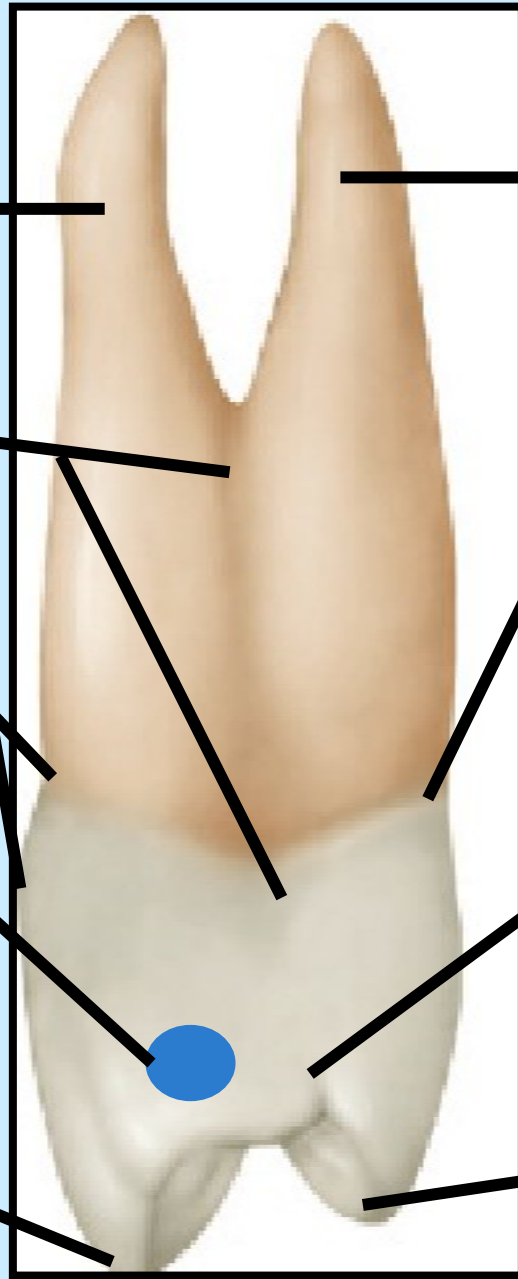
LINGUAL ROOT

CERVICAL LINE

**MESIAL
MARGINAL
DEVELOPMENTAL
GROOVE**

LINGUAL CUSP

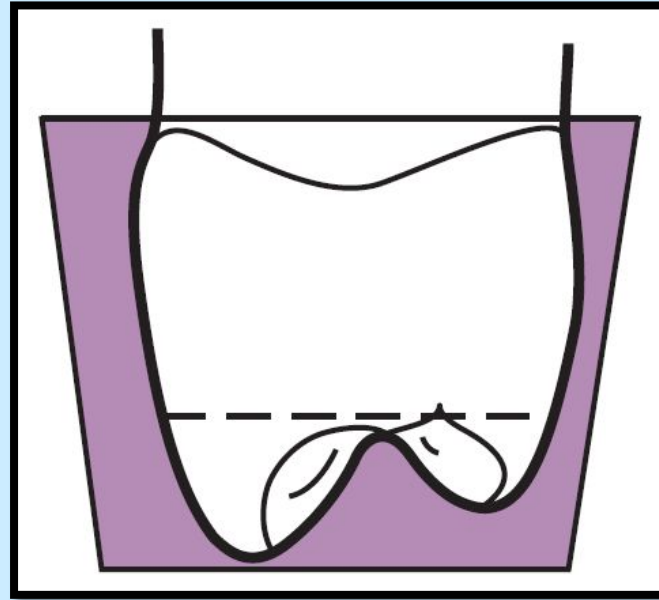
**MESIAL
ASPECT**



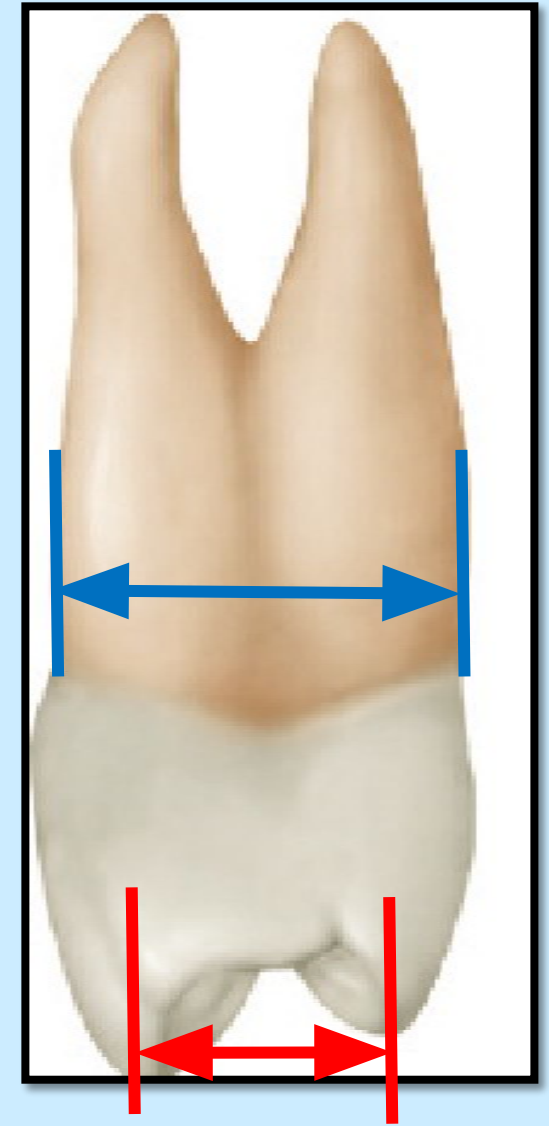
Mesial surface shape =
roughly *trapezoidal*.

***Characteristic of all
posterior teeth:***

Cusp tips are well within
confines of root trunk.



**MESIAL
ASPECT**



Measurement from buccal to lingual cusp tip <
BL measurement of the crown/root at its
cervical portion.

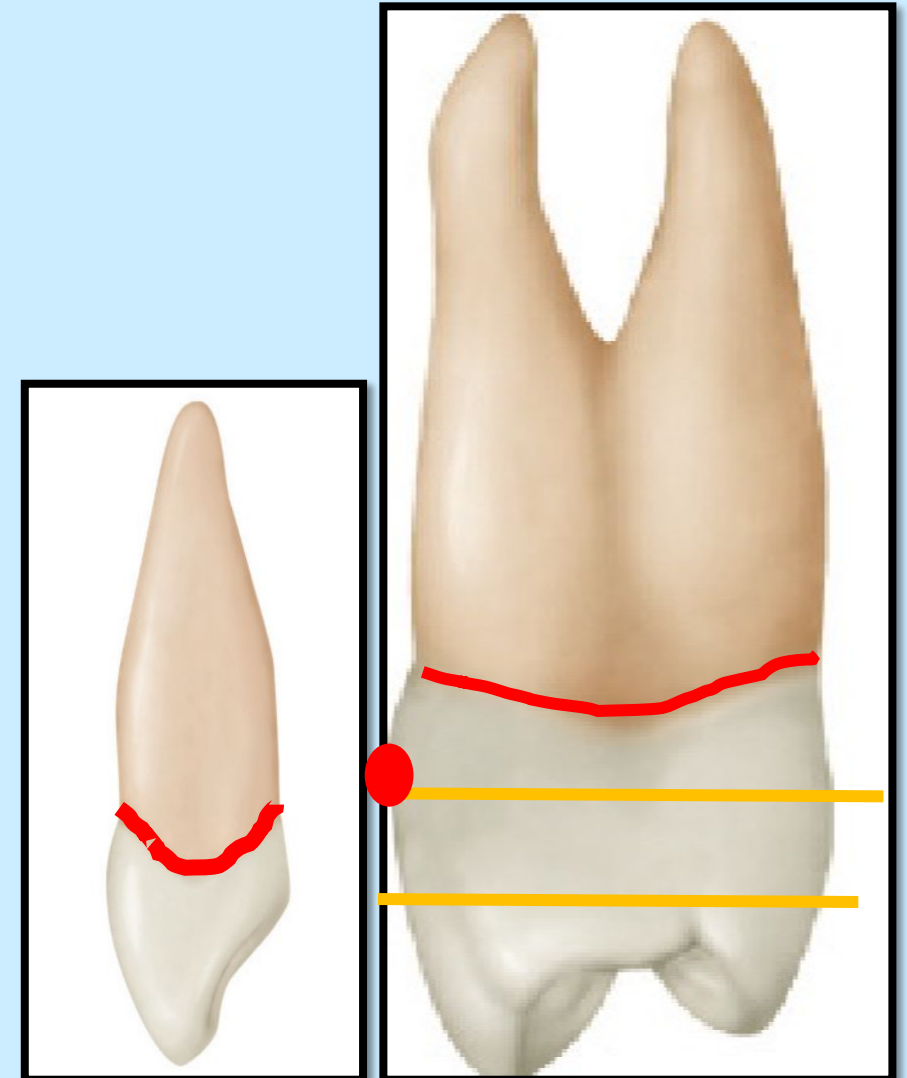
Cervical line.

- Cervical line curvature occlusally on mesial aspect is 1 mm **less** than that on mesial of any anterior tooth.

Buccal crest of contour (cervical ridge)

Located at junc. of cervical & middle 3rd OR Cervical 3rd.

MESIAL ASPECT

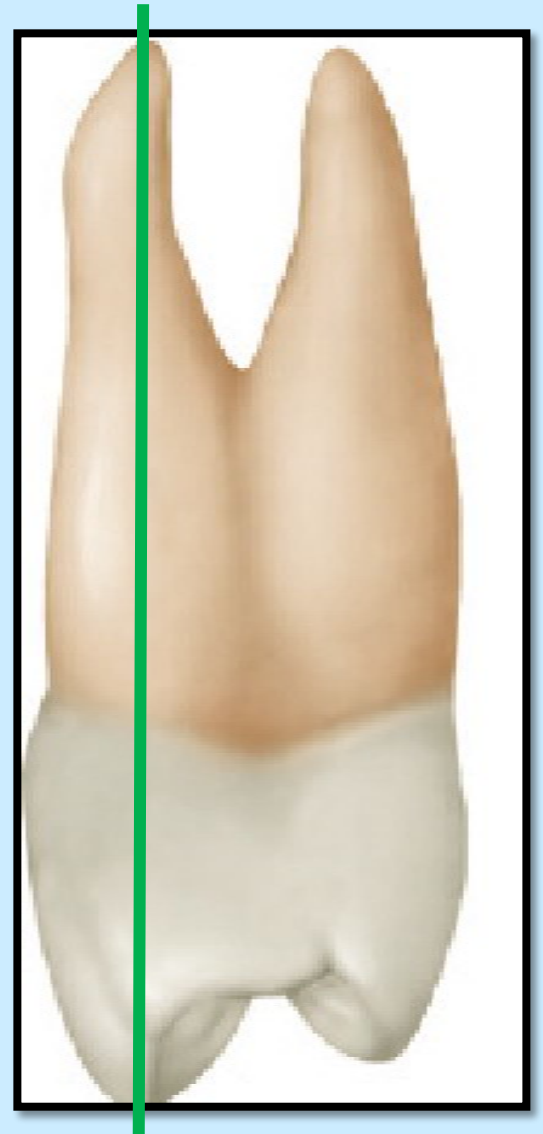


MESIAL ASPECT

Buccal Outline:

- Curved overall
- Less convex from crest of contour to B cusp tip

B cusp tip in line with B root centre



Lingual crest of contour

- Middle 3rd

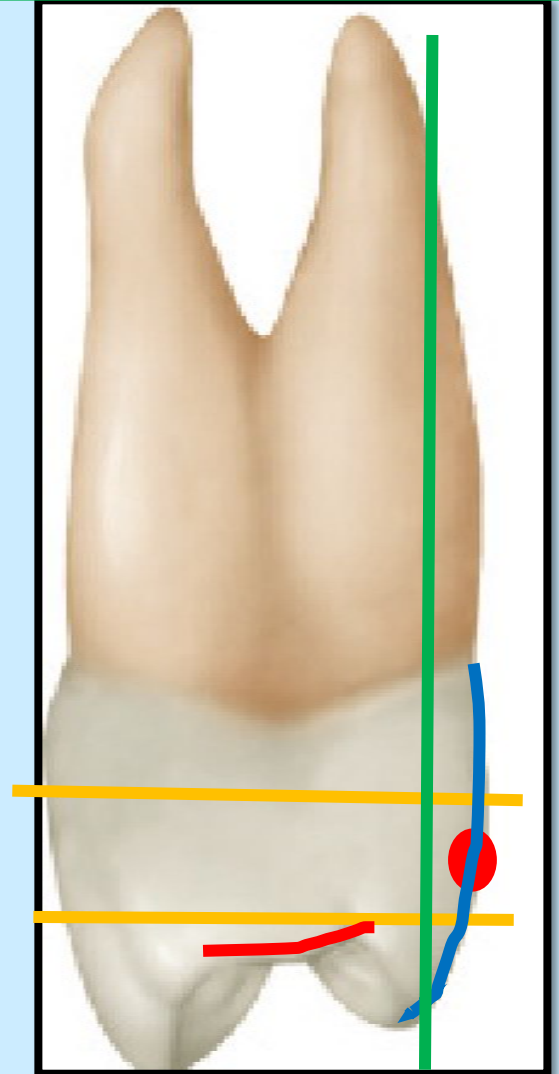
Lingual Outline:

- Smoothly curved from cervical line to L cusp tip.

Location of Mesial Marginal Ridge

- Junction of middle & occlusal 3rd.

L cusp tip in line with L border of L root.



**MESIAL
ASPECT**

Class Traits & Key Considerations:

Lingual height of contour is midway between the cervical line and cusp tip.

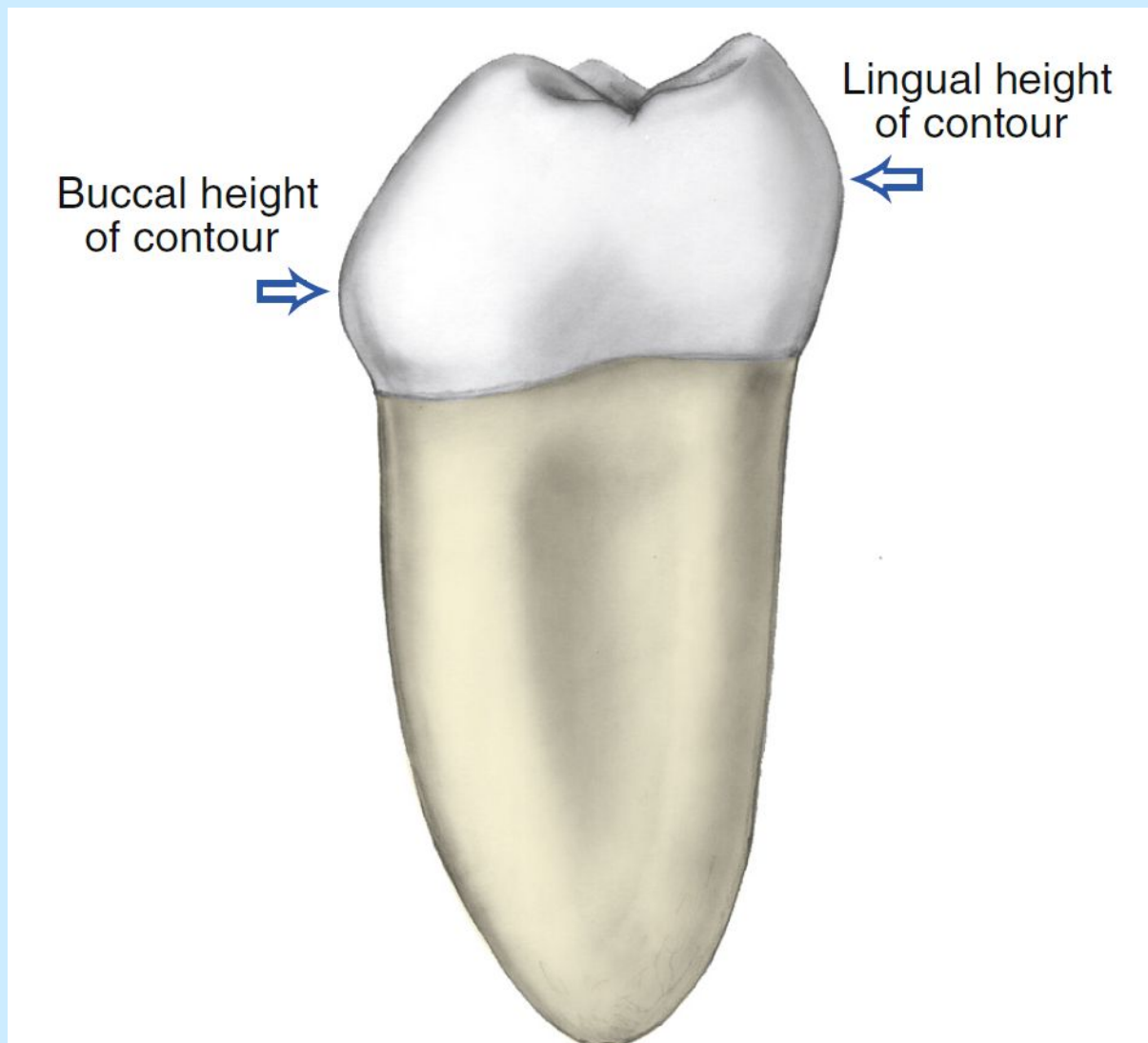
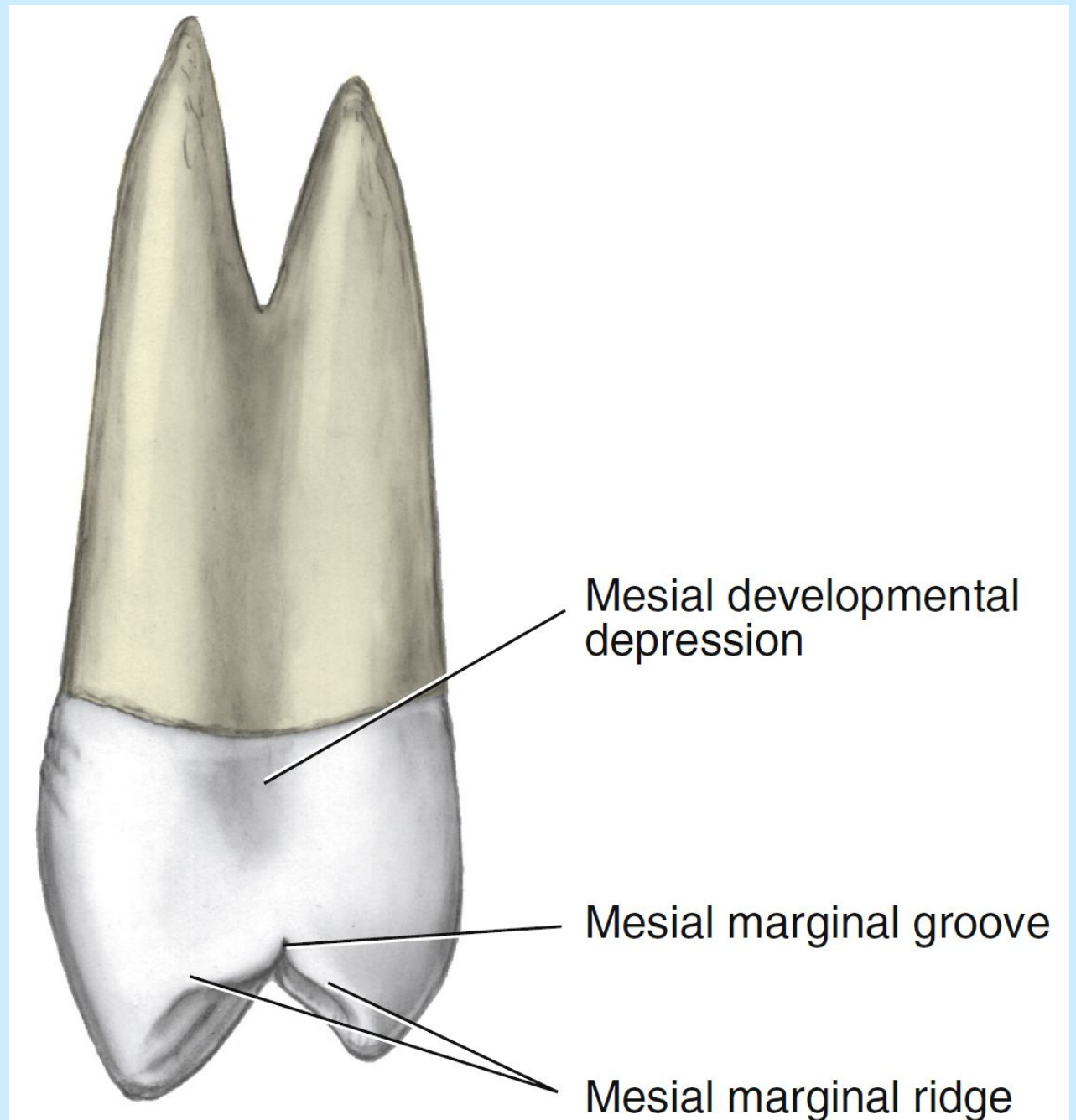
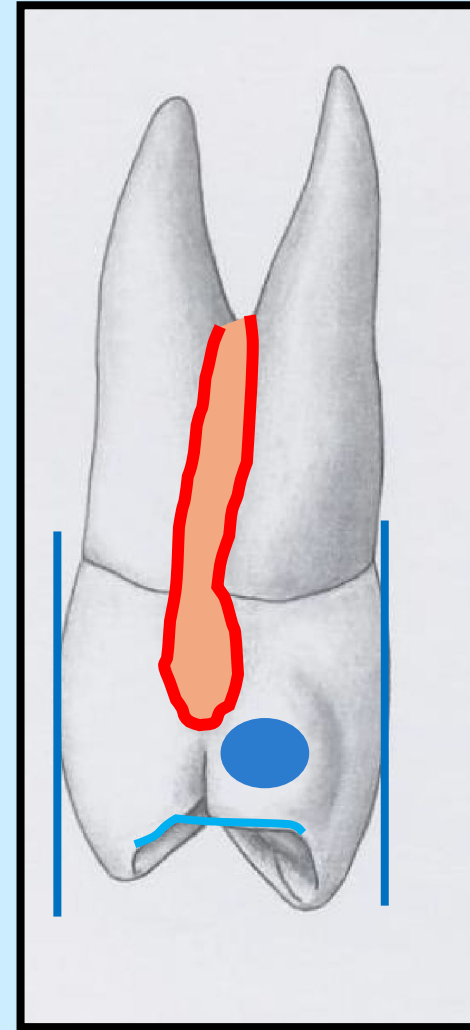


FIGURE 17-14 Mesial features of the permanent maxillary right first premolar.

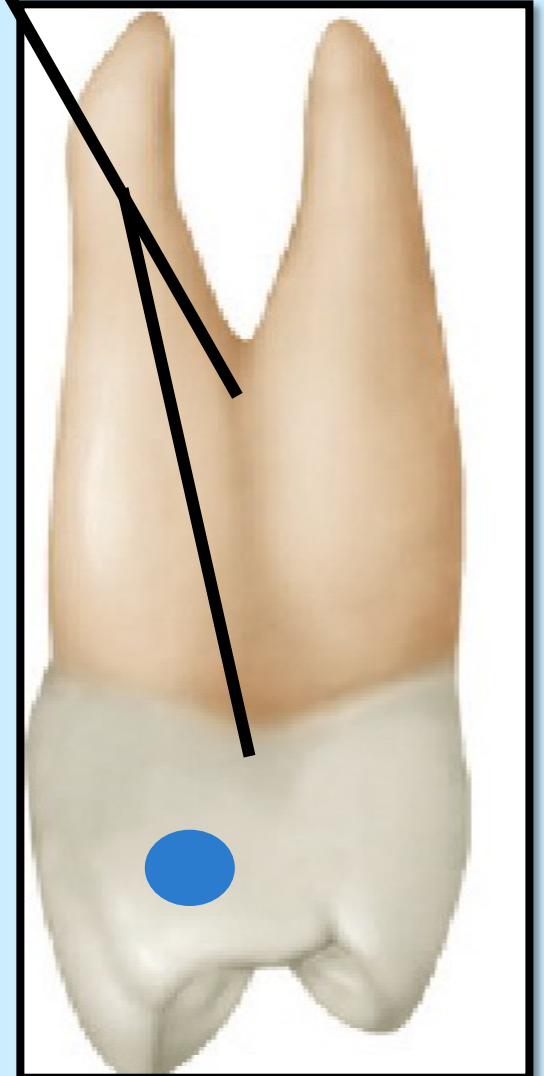


- **Mesial Developmental Depression** Located immediately **cervical to MCA**, **centered** on the M surface and **bordered by the MB & ML line angles**.
- This mesial concavity continues apically beyond the cervical line, joins a **Deep Developmental Depression** between the roots, and ends at the root bifurcation.

**MESIAL
DEVELOPMENTAL
DEPRESSION**

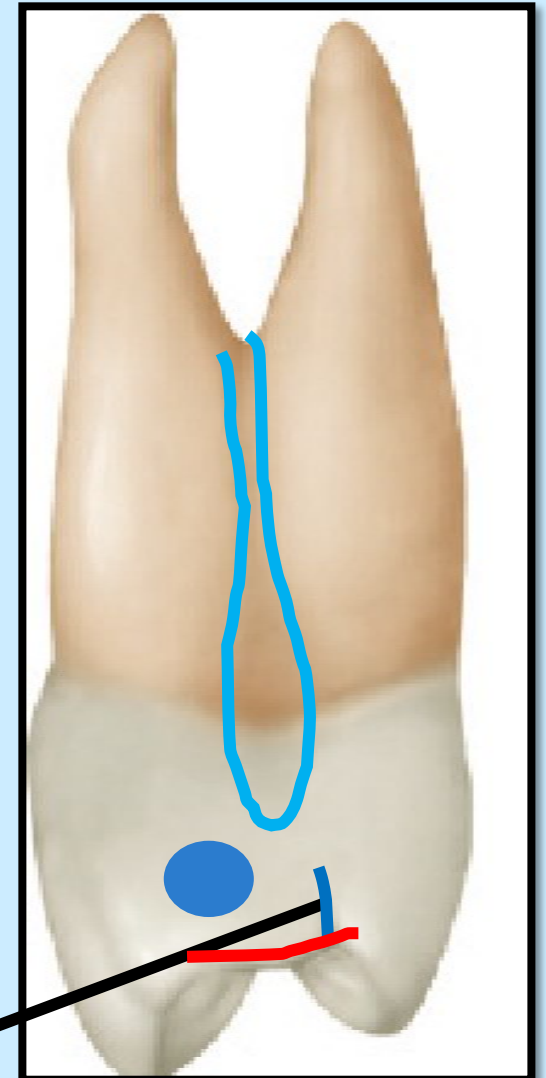


**MESIAL
ASPECT**



- **Mesial Marginal Developmental Groove** is well-defined in the enamel of the mesial marginal ridge.
- Well aligned with M root **developmental depression** but not connected.
- **continuous with the central groove** of the occlusal surface of the crown, **crossing the marginal ridge** immediately **L to the MCA** & **terminating** a short distance **cervical to the M marginal ridge** on the M surface

**MESIAL
ASPECT**



**MESIAL MARGINAL
DEVELOPMENTAL GROOVE**

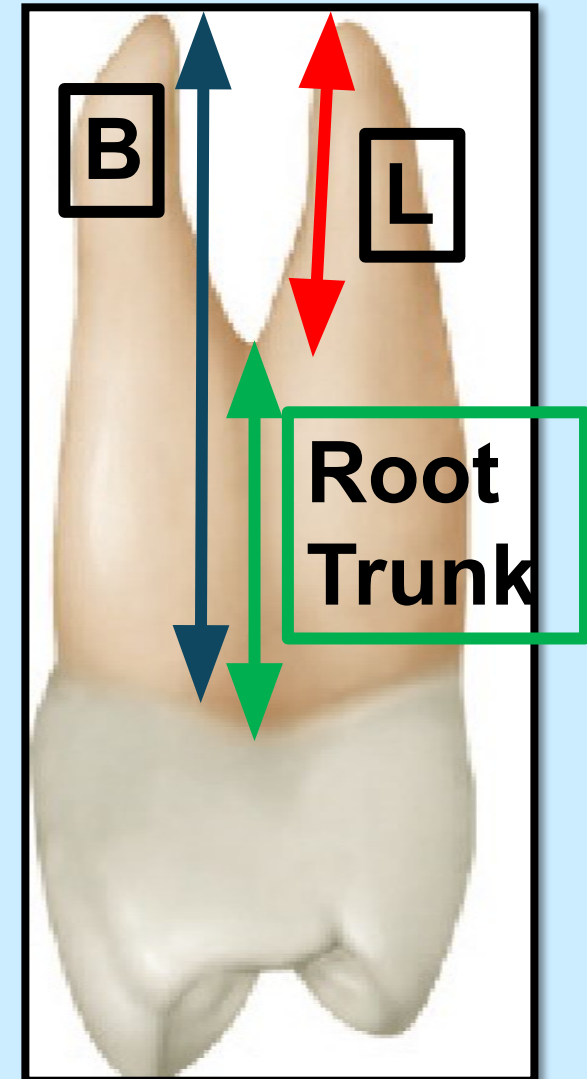
Root trunk length & Bifurcation

- Root is bifurcated for $\frac{1}{2}$ its total length.
- Root trunk is long on this tooth, making up about $\frac{1}{2}$ of the root length.

Buccal And Lingual Roots

Considerable curvature apical to middle 3rds. B or L inclination

MESIAL ASPECT



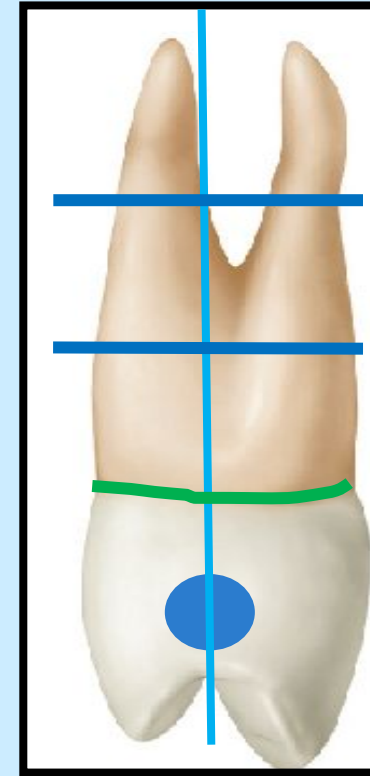
DISTAL ASPECT

Smooth convexity overall, except flattened arc cervical to Distal contact area & B to center of D surface.

Less or almost straight cervical line,

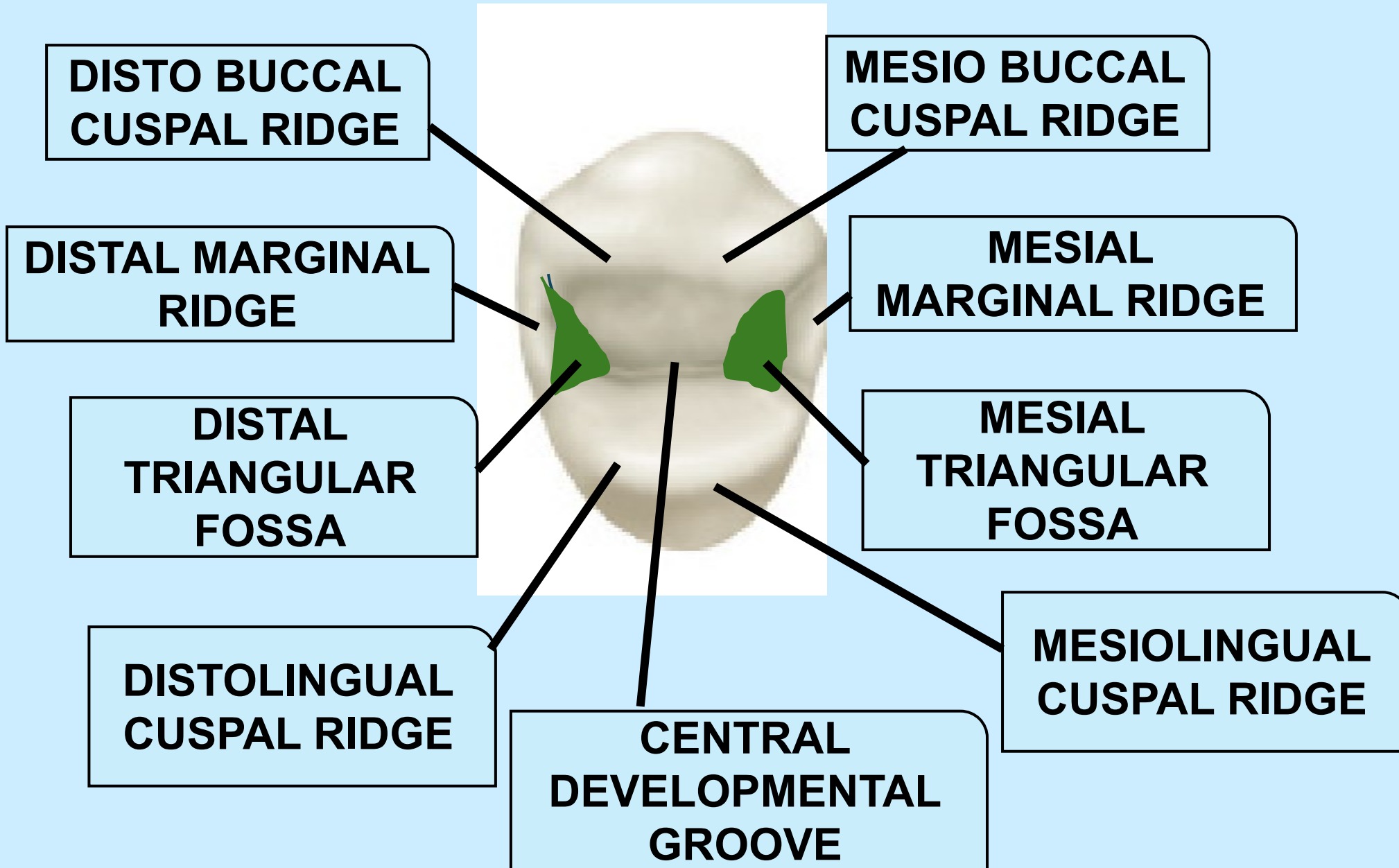
Absent or shallow or insignificant developmental groove.

DIFFERENCES FROM MESIAL SURFACE



**DISTAL
ASPECT**

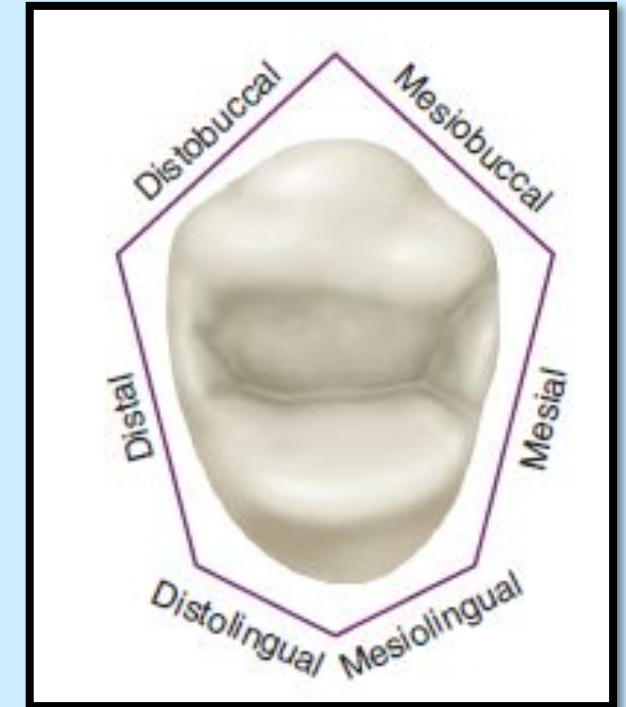
OCCLUSAL ASPECT



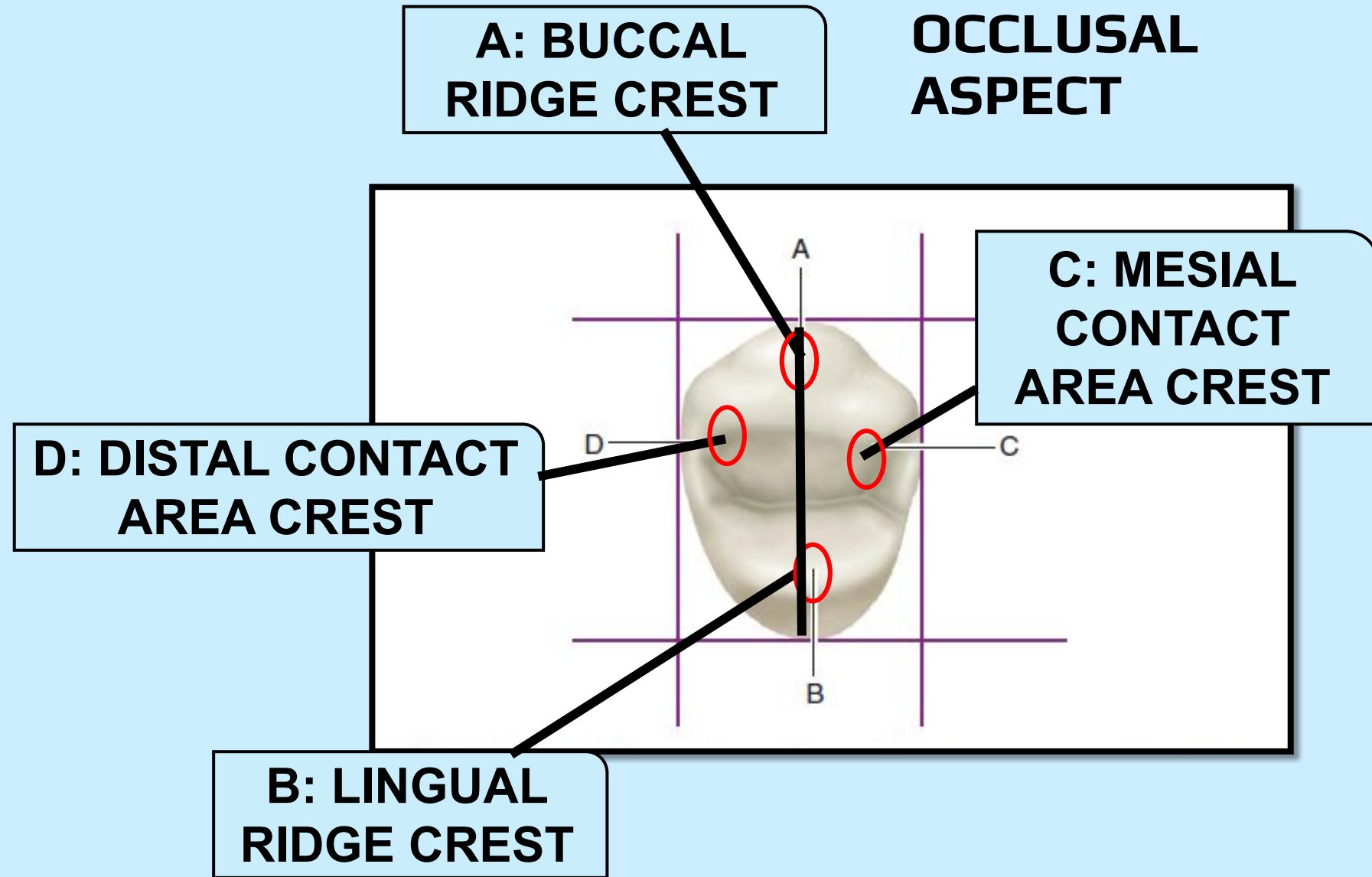
OCCLUSAL ASPECT

roughly a **six-sided or hexagonal figure**

- hexagonal figure $\neq$ **equilateral**.
- The **2 buccal sides** are **nearly equal**
- **M side** < **D side**
- **ML side** < **DL side**

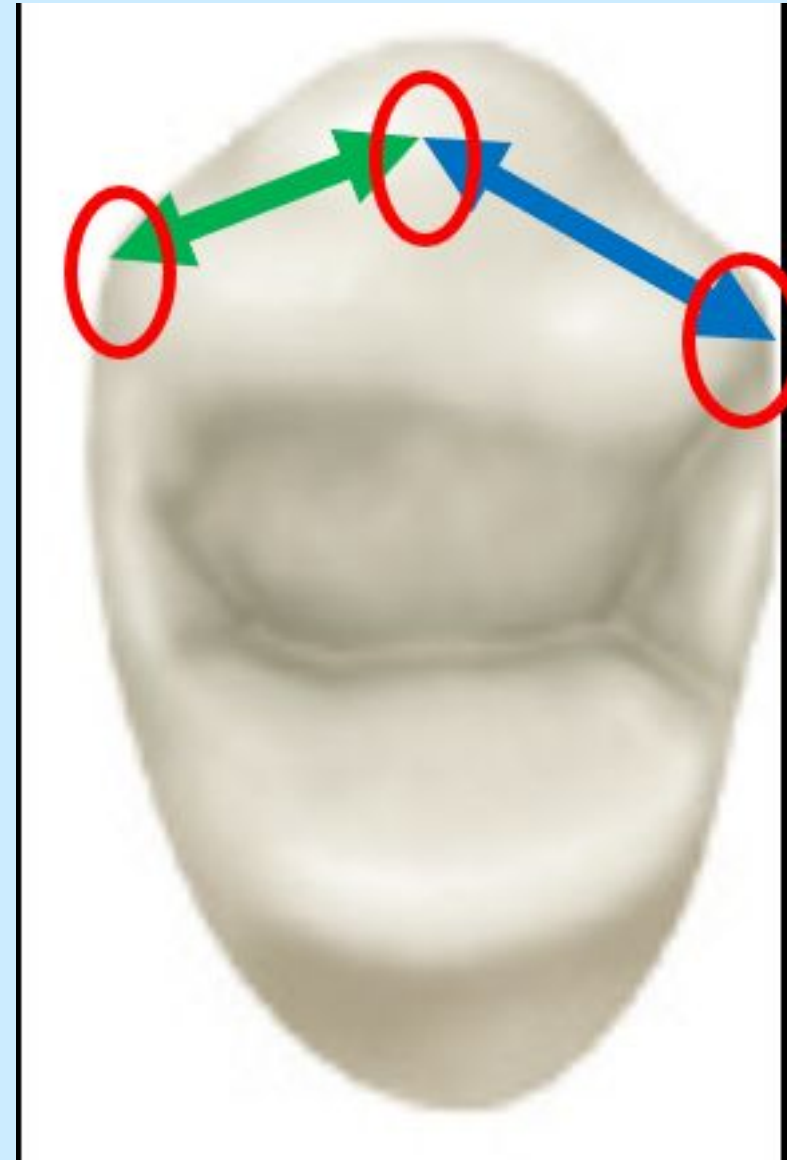


- **Distal Contact** somewhat **B** to **Mesial Contact Area**
- **Buccal ridge crest** somewhat **D** to **lingual ridge crest**.



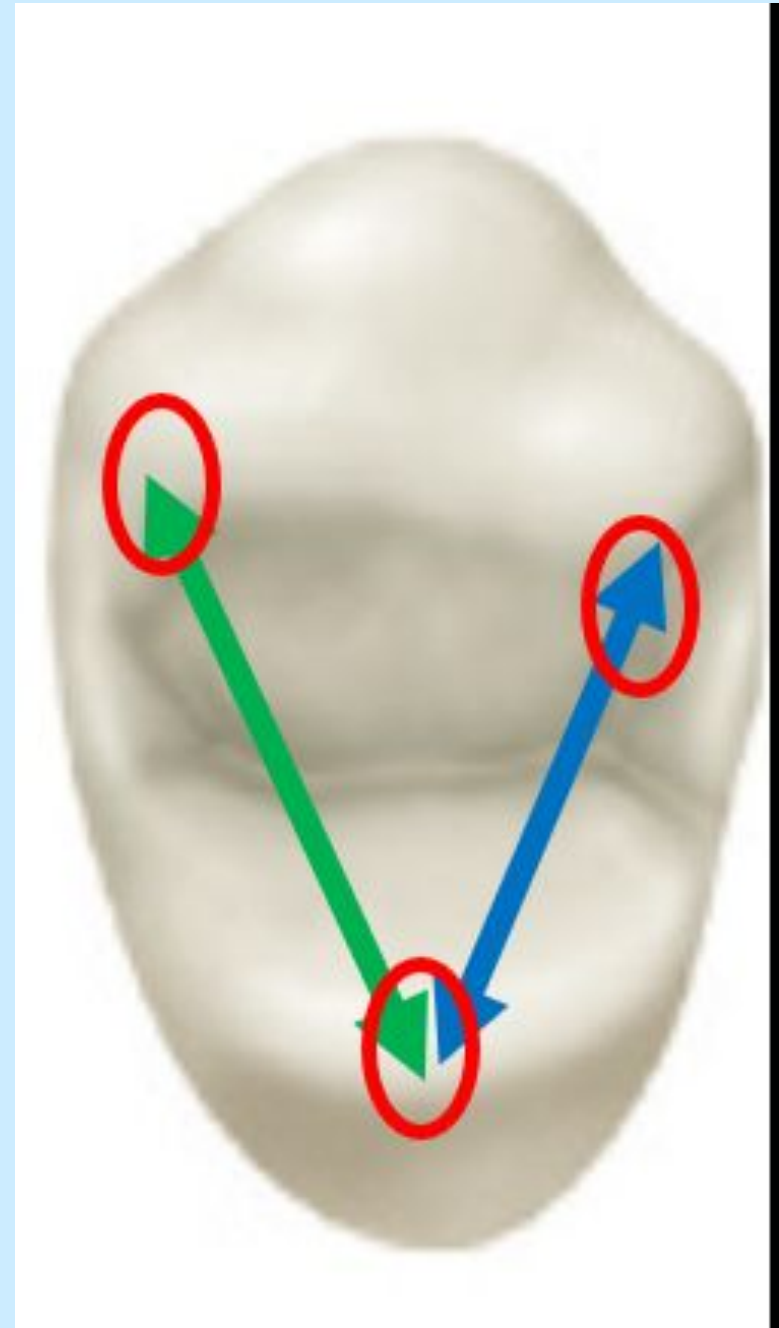
**Buccal cusp tip →
mesial crest is longer
than Buccal cusp tip
→ distal crest.**

- **The buccal cusp tip is
displaced slightly
distally.**



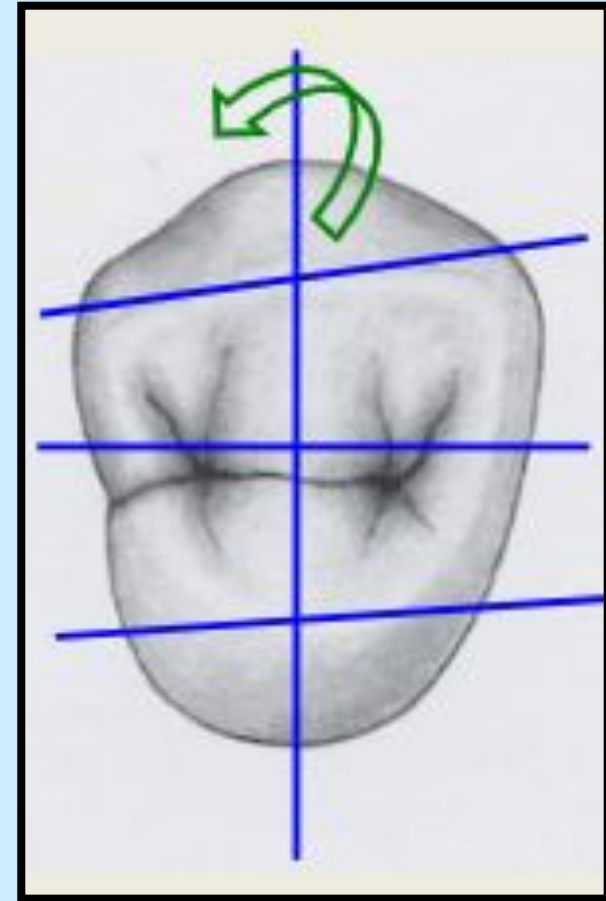
The distance from the **mesial crest to the lingual cusp tip** is less than that from the distal crest.

- The **lingual cusp tip is displaced slightly mesially.**



The short M vs. D surface measurement and position of the B & L cusp tips produce a “twisted” or “rotated” appearance around its long axis.

OCCLUSAL ASPECT



MB & DB Cuspal Ridges:

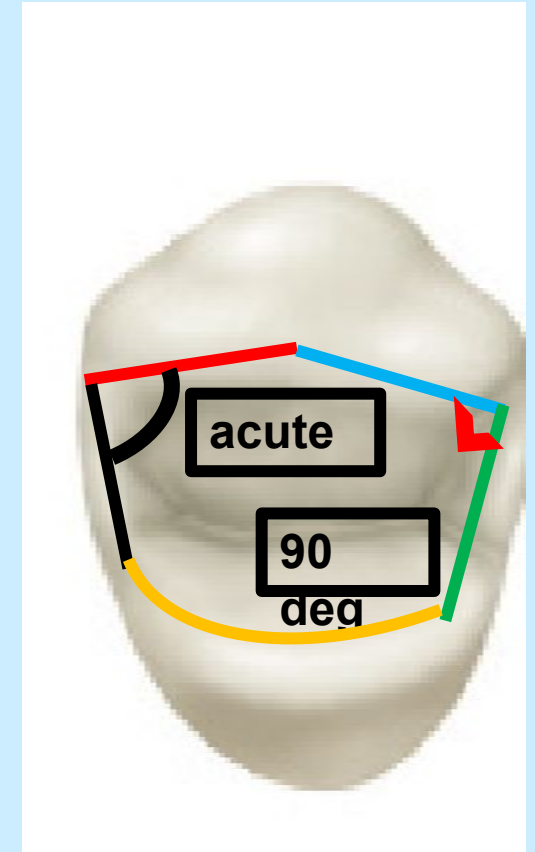
- Well aligned

Angle b/w MB cuspal ridge & M marginal ridge:
90 degrees

Angle b/w DB cuspal ridge & D marginal ridge:
• acute

- ML & DL cusp ridges ***confluent with M & D marginal ridges***
- ***Curved & semicircular***

OCCLUSAL ASPECT

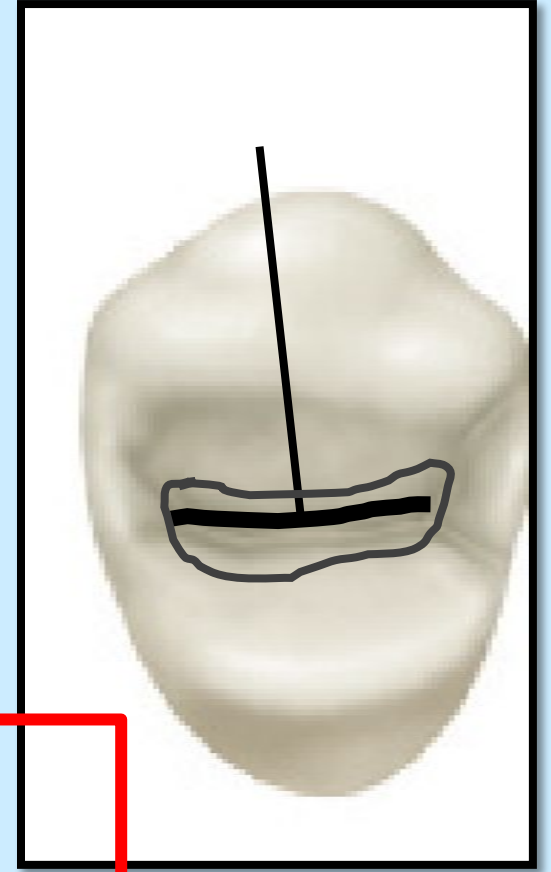


More B than L surface visible when occlusal aspect is viewed.

OCCLUSAL ASPECT

Central Developmental Groove:

- **well-defined** & divides the surface evenly BL.
- It is located at **bottom of central sulcus** of the occlusal surface & extending from a point just **M to D marginal ridge towards M marginal ridge**, where it joins the **M marginal development groove**.



No supplemental grooves = smooth surface

Mesial marginal developmental groove:

- Is connected with CDG.
- crosses the mesial marginal ridge and ends on the mesial surface of the crown

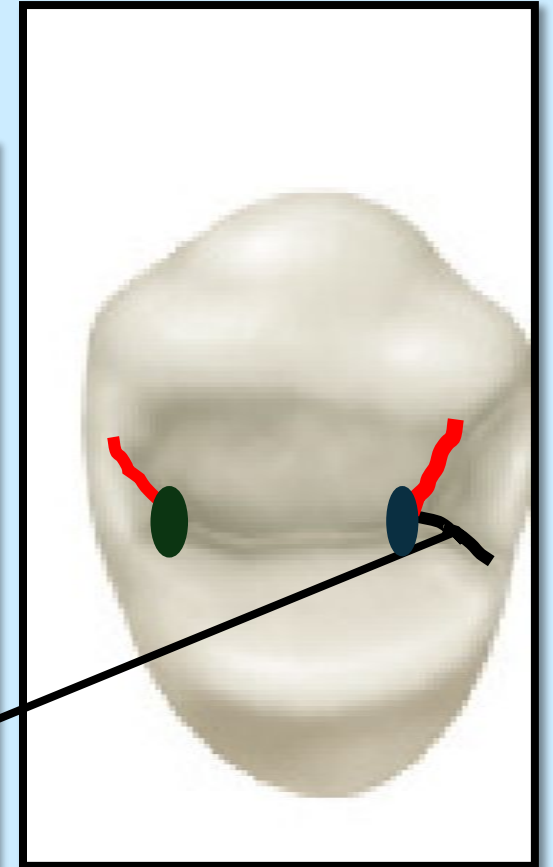
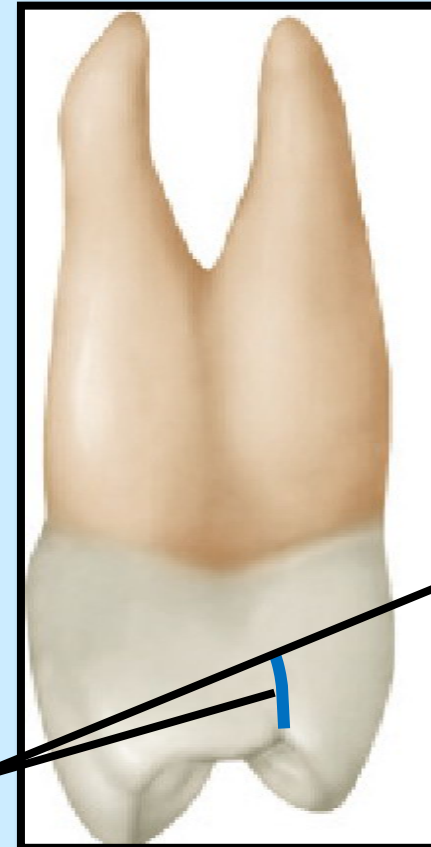
MB & DB developmental groove:

join the CG just inside the M & D marginal ridges.

M & D pits present at junctions of M & D developmental grooves with CG respectively

MESIAL MARGINAL DEVELOPMENTAL GROOVE

**OCCLUSAL
ASPECT**



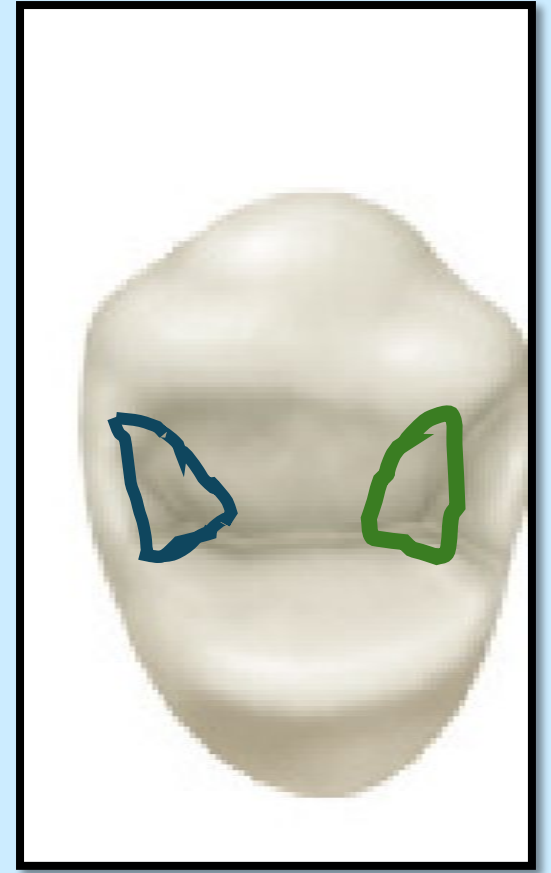
M Triangular Fossa:

Just D to the M marginal ridge, the triangular depression that harbors the MB developmental groove is called the M triangular fossa.

D Triangular Fossa:

The depression in the occlusal surface, just M to the D marginal ridge, is called the D triangular fossa.

OCCLUSAL ASPECT



B triangular ridge :

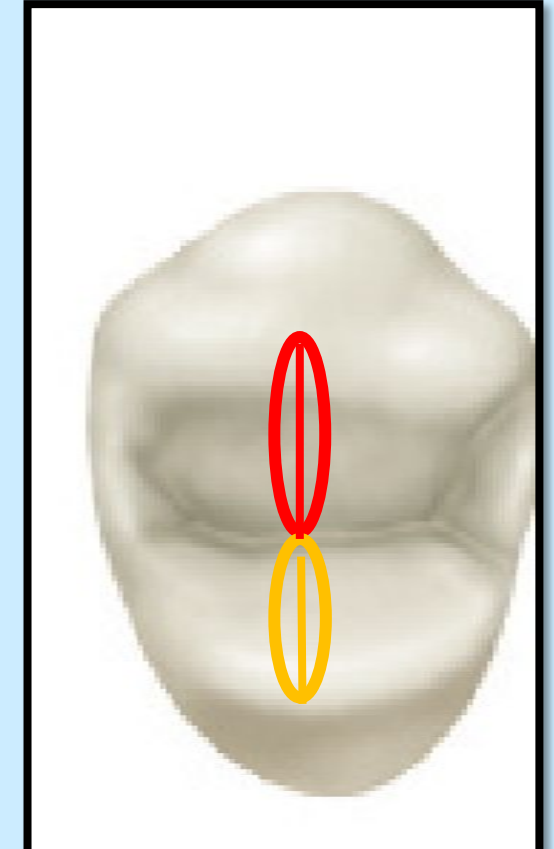
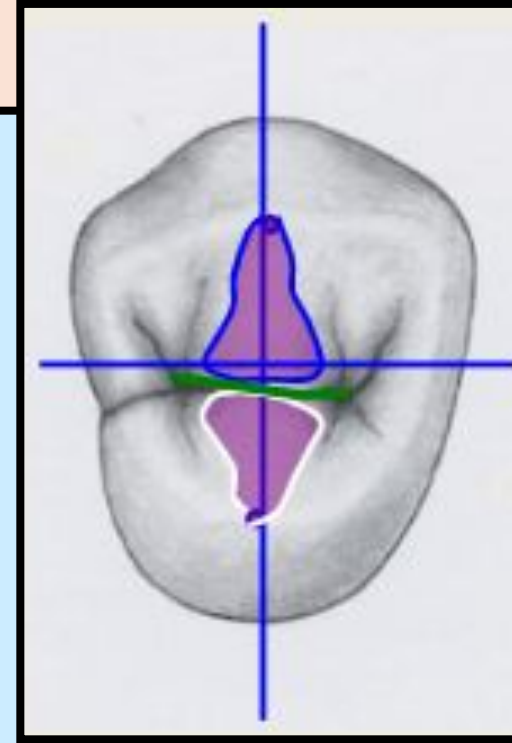
The B triangular ridge of the B cusp is prominent, arising near the center of the CG & converging with the tip of the B cusp.

L triangular ridge:

The L triangular ridge is less prominent; it also arises near the center of the CG & converges with the tip of the L cusp.

Transverse Ridge
separated by deep CG

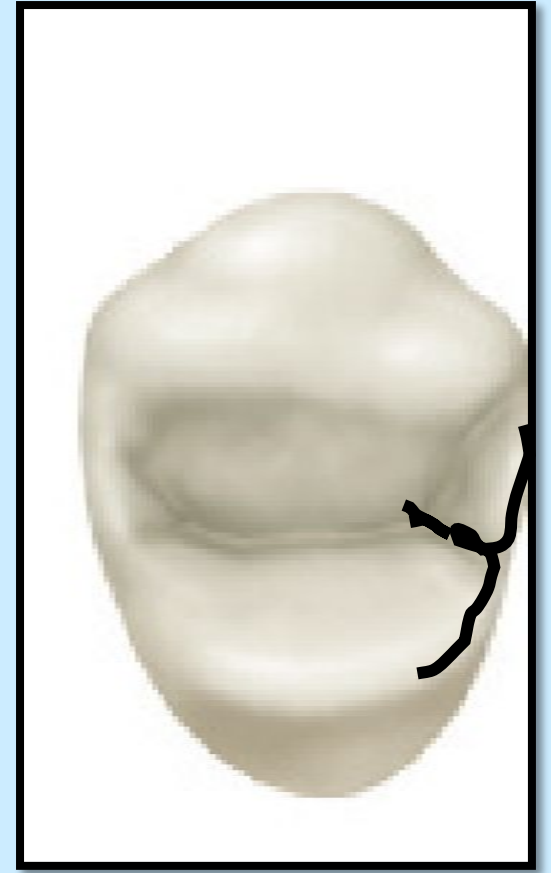
OCCLUSAL ASPECT



L cusp is pointed more sharply than B cusp.

OCCLUSAL ASPECT

The **mesial outline** reflects the contour of a **reverse figure “3”** due to the **presence of the mesial marginal ridge developmental groove**.



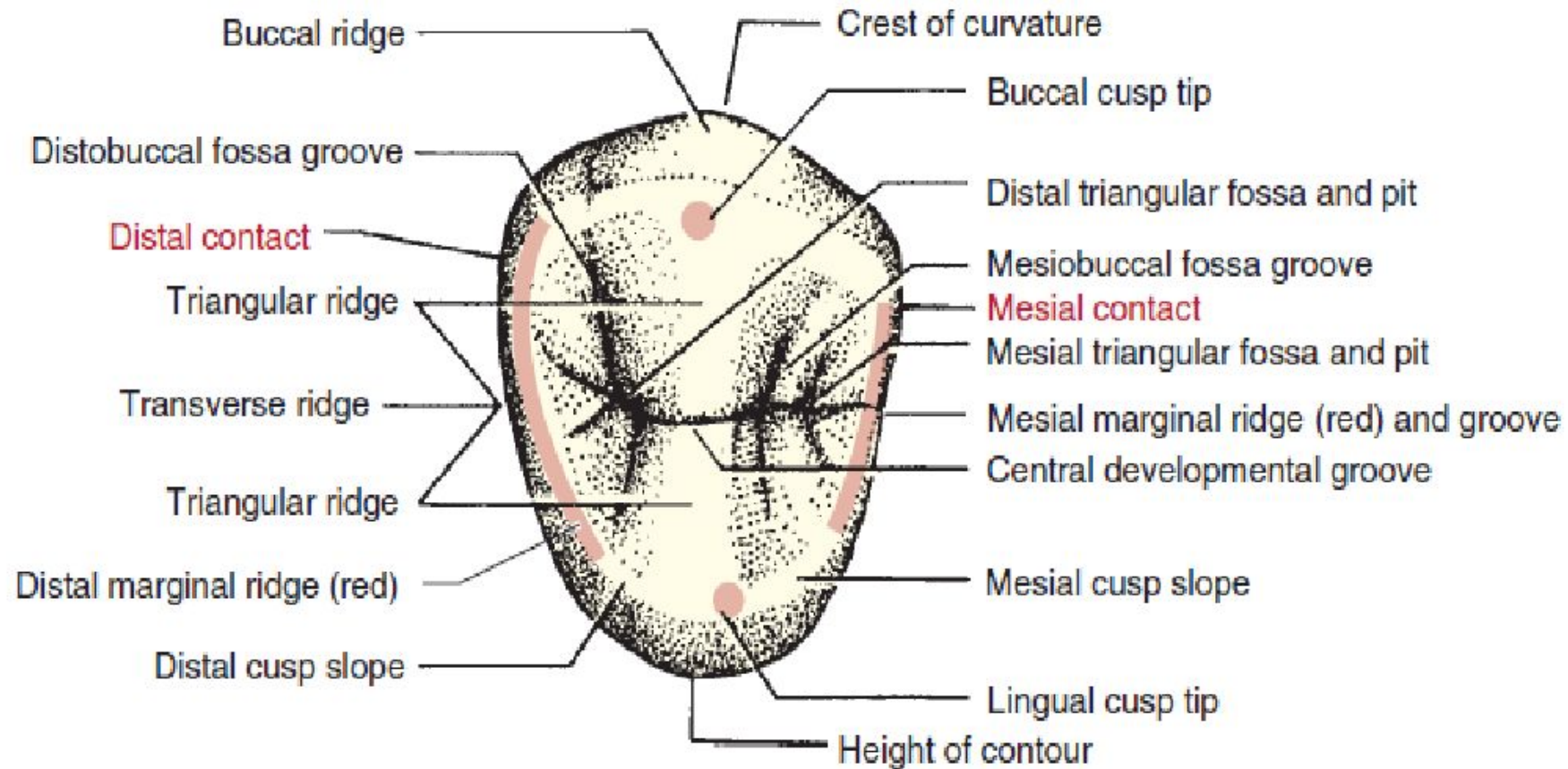


FIGURE 4-10.

Maxillary right first premolar, occlusal surface, with anatomic structures that contribute to its **asymmetry**. Notice the proximal contact locations (the distal contact of this tooth is more buccal than the mesial), the relative length of the marginal ridges (mesial is shorter), and the location of the cusp tips where the buccal tip is distal to the middle, making the mesial cusp ridge of the buccal cusp longer than the distal cusp ridge (UNIQUE to maxillary first premolars) and the lingual cusp tip is mesial to the middle.



Thank you